

# A Three-Rationality Hierarchy for Composite Systems: Local, Order-Preserving, and Time-Preserving Optimality under Scale Separation

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## Abstract

We formulate a hierarchical theory of rational action in finite-dimensional composite quantum systems, identifying three coexisting rationalities operating on separated time-scales: local rationality for short-scale node-level optimization, order-preserving rationality for intermediate-scale preservation of inter-node coupling structure, and time-preserving rationality for long-scale conservation of future reachability. We give operational definitions for each, with particular emphasis on (i) the admissible-order set  $\mathcal{O}_{\text{adm}}$  defined through Cheeger conductance and a separation of structural and dynamic conditions, (ii) the admissible-flow cone  $\mathcal{K}_{\text{adm}}$  characterizing non-domination of subsystem coalitions, (iii) the irreversibility cost  $I_{\text{irr}}$  defined as deviation from the unitary orbit, and (iv) the future volume  $V_{\text{future}}$  defined on Hilbert–Schmidt measure. Through singular perturbation analysis we prove a Tikhonov-type slow-manifold convergence and establish two main results. The Feasibility Inclusion Theorem shows that admissible initial conditions of a two-layer reduced system extend to the full three-layer system. The Buffer Stability Theorem shows that the three-layer hierarchy strictly dominates a two-layer system in terms of accumulated PFH-constraint violation when initial conditions lie off the slow manifold. We illustrate the results with a numerical experiment on a three-node finite system. The framework abstracts away domain-specific content (no political, biological, or organizational interpretations are invoked) so that the same law applies uniformly across complex composite systems.

## Contents

|          |                                                                |          |
|----------|----------------------------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                                            | <b>3</b> |
| <b>2</b> | <b>Preliminaries and Notation</b>                              | <b>4</b> |
| 2.1      | Composite system . . . . .                                     | 4        |
| 2.2      | Coupling structure . . . . .                                   | 4        |
| 2.3      | Control and dynamics . . . . .                                 | 4        |
| <b>3</b> | <b>Three Time-Scales and the Hierarchical Setting</b>          | <b>5</b> |
| 3.1      | Why three layers . . . . .                                     | 5        |
| <b>4</b> | <b>Local Rationality</b>                                       | <b>5</b> |
| 4.1      | Definition . . . . .                                           | 5        |
| 4.2      | Externality . . . . .                                          | 6        |
| 4.3      | Decoupling of fast state dynamics from coupling rate . . . . . | 6        |

|           |                                                                 |           |
|-----------|-----------------------------------------------------------------|-----------|
| <b>5</b>  | <b>Order-Preserving Rationality</b>                             | <b>6</b>  |
| 5.1       | Structural admissibility . . . . .                              | 6         |
| 5.2       | Dynamic admissibility . . . . .                                 | 7         |
| 5.3       | Combined admissibility . . . . .                                | 7         |
| 5.4       | Admissible-flow cone . . . . .                                  | 7         |
| 5.5       | Order-preserving cost . . . . .                                 | 8         |
| 5.6       | Geometric properties of $\mathcal{K}_{\text{adm}}$ . . . . .    | 9         |
| 5.7       | Conservativeness of the convex relaxation . . . . .             | 9         |
| <b>6</b>  | <b>Time-Preserving Rationality</b>                              | <b>10</b> |
| 6.1       | Reachable and admissible futures . . . . .                      | 10        |
| 6.2       | Future volume . . . . .                                         | 10        |
| 6.3       | Irreversibility . . . . .                                       | 11        |
| 6.4       | Time-preserving cost . . . . .                                  | 11        |
| <b>7</b>  | <b>Hierarchical Optimization and Singular Perturbation</b>      | <b>12</b> |
| 7.1       | Three-layer dynamics . . . . .                                  | 12        |
| 7.2       | Two-layer reduction . . . . .                                   | 12        |
| 7.3       | Tikhonov-type slow-manifold lemma . . . . .                     | 12        |
| <b>8</b>  | <b>Buffer Stability Theorem</b>                                 | <b>13</b> |
| 8.1       | Feasibility sets . . . . .                                      | 13        |
| 8.2       | Strict admissibility and the lift theorem . . . . .             | 14        |
| 8.3       | Buffer Stability under Barrier-Regularized Order Flow . . . . . | 15        |
| 8.4       | Corollaries of the Buffer Stability Theorem . . . . .           | 18        |
| <b>9</b>  | <b>Numerical Illustration</b>                                   | <b>19</b> |
| 9.1       | Setup . . . . .                                                 | 19        |
| 9.2       | Results . . . . .                                               | 19        |
| 9.3       | Quantitative tests . . . . .                                    | 19        |
| <b>10</b> | <b>Discussion</b>                                               | <b>21</b> |
| 10.1      | Relation to existing frameworks . . . . .                       | 21        |
| 10.2      | Buffer principle . . . . .                                      | 22        |
| 10.3      | Mathematical placement . . . . .                                | 22        |
| 10.4      | Buffer principle vs. Lyapunov stability . . . . .               | 23        |
| 10.5      | Limitations and Paper 2 . . . . .                               | 23        |
| <b>11</b> | <b>The Three-Rationality Hierarchy as an Abstract Principle</b> | <b>24</b> |
| 11.1      | Compatibility with the Photon Reactor framework . . . . .       | 24        |
| 11.2      | Three-rationality in summary . . . . .                          | 25        |
| <b>12</b> | <b>Conclusion</b>                                               | <b>26</b> |
| <b>A</b>  | <b>Present-cost and predictive-uncertainty terms</b>            | <b>26</b> |
| <b>B</b>  | <b>Cheeger inequality and verification</b>                      | <b>26</b> |
| <b>C</b>  | <b>Auxiliary lemmas used in the proofs</b>                      | <b>27</b> |
| <b>D</b>  | <b>Numerical implementation</b>                                 | <b>28</b> |
| <b>E</b>  | <b>Notation summary</b>                                         | <b>29</b> |

# 1 Introduction

A composite system in equilibrium is, almost by definition, a system in which competing pressures are simultaneously satisfied. The pressures, however, do not all live at the same time-scale. The rationality that governs whether a node maximizes its own utility now is not the same as the rationality that governs whether the inter-node coupling structure remains predictable this season, which is in turn not the same as the rationality that governs whether the system retains future evolutionary capacity across generations. These are three rationalities, not one.

Existing optimization frameworks tend to collapse this distinction. Reinforcement learning maximizes a single (possibly time-discounted) reward. Mean-field game theory equilibrates over a single agent population. Even multi-objective optimization frameworks treat the three pressures as parallel, weighted by a single Lagrangian, which is mathematically incorrect when the time-scales are separated.

This paper develops the formal framework that respects the three-fold separation. The main contributions are:

1. A precise operational definition of the admissible-order set  $\mathcal{O}_{\text{adm}}$ , which separates structural conditions (minimum connectivity and Cheeger conductance) from dynamic conditions (future reachability and constraint compatibility), thereby distinguishing genuine order from coalitional capture without invoking any extra-mathematical content (Section 5).
2. A precise operational definition of the admissible-flow cone  $\mathcal{K}_{\text{adm}}$ , capturing in convex form the requirement that no proper subsystem may simultaneously strengthen its internal couplings while severing its external ones (Section 5).
3. Operational definitions of the irreversibility cost  $I_{\text{irr}}$  as the integrated trace-distance from the unitary orbit, and the future volume  $V_{\text{future}}$  as a Hilbert–Schmidt measure of the intersection of the reachable set with the admissible-future set (Section 6).
4. A singular perturbation reformulation in which the three rationalities arise as fast, intermediate, and slow layers, with a Tikhonov-type lemma ensuring slow-manifold convergence (Section 7).
5. Two main theorems: Feasibility Inclusion (Theorem 8.2) and Buffer Stability (Theorem 8.10), the latter being the formal expression of the claim that the three-layer hierarchy dominates the two-layer reduction in terms of PFH-violation suppression (Section 8).
6. A numerical illustration on a three-node finite-dimensional system (Section 9).

The framework is deliberately abstract. Although the three-rationality intuition is recoverable from a wide variety of concrete settings—neural systems, ecological networks, multi-agent reinforcement learning, multi-AI coordination, and yes, social organization—we make no domain-specific commitment in this paper. Domain interpretations are deferred to companion work.

## 2 Preliminaries and Notation

### 2.1 Composite system

Let  $V = \{1, 2, \dots, N\}$  index a finite collection of nodes. Each node  $i$  carries a finite-dimensional Hilbert space  $\mathcal{H}_i$  of dimension  $d_i$ ; the composite Hilbert space is

$$\mathcal{H}_{\text{sys}} = \bigotimes_{i=1}^N \mathcal{H}_i, \quad \dim \mathcal{H}_{\text{sys}} = \prod_i d_i =: D. \quad (1)$$

The space of density operators is  $\mathcal{D}(\mathcal{H}_{\text{sys}}) = \{\rho \in \mathcal{B}(\mathcal{H}_{\text{sys}}) : \rho = \rho^\dagger, \rho \geq 0, \text{Tr } \rho = 1\}$ . We use the Hilbert-Schmidt inner product  $\langle A, B \rangle_{\text{HS}} = \text{Tr}(A^\dagger B)$  and norm  $\|A\|_{\text{HS}} = \sqrt{\text{Tr}(A^\dagger A)}$ . The trace norm is  $\|A\|_1 = \text{Tr}|A|$ . The reduced state at node  $i$  is  $\rho_i = \text{Tr}_{\bar{i}}(\rho)$ , where  $\text{Tr}_{\bar{i}}$  denotes the partial trace over  $\bigotimes_{j \neq i} \mathcal{H}_j$ .

### 2.2 Coupling structure

The order between nodes is encoded by a symmetric, off-diagonal, non-negative coupling matrix

$$\Gamma = (\gamma_{ij})_{i,j=1}^N, \quad \gamma_{ij} = \gamma_{ji} \geq 0, \quad \gamma_{ii} = 0. \quad (2)$$

We write  $\Gamma \in \mathcal{C}_N := \{\Gamma \in \mathbb{R}^{N \times N} : \text{symmetric, non-negative, zero diagonal}\}$ . Each  $\gamma_{ij}$  is interpretable as the strength of an exchange/communication/correlation channel between  $i$  and  $j$  but no specific interpretation is invoked. The weighted degree at node  $i$  is

$$d_i(\Gamma) = \sum_{j \neq i} \gamma_{ij}. \quad (3)$$

For  $S \subseteq V$ , the cut and volume of  $S$  are

$$\text{cut}_\Gamma(S, S^c) = \sum_{i \in S, j \notin S} \gamma_{ij}, \quad \text{vol}_\Gamma(S) = \sum_{i \in S} d_i(\Gamma). \quad (4)$$

The Cheeger conductance of  $S$  is

$$\Phi_\Gamma(S) = \frac{\text{cut}_\Gamma(S, S^c)}{\min\{\text{vol}_\Gamma(S), \text{vol}_\Gamma(S^c)\}}, \quad \emptyset \subsetneq S \subsetneq V. \quad (5)$$

### 2.3 Control and dynamics

The system is driven by a control protocol  $u(\cdot) \in \mathcal{U}$  whose admissible class is fixed throughout. The dynamics of the state are governed by a parameterized Lindblad-type generator  $\mathcal{L}_{u,\Gamma,\Theta}$ :

$$\dot{\rho} = \mathcal{L}_{u,\Gamma,\Theta}[\rho] = -i[H_{u,\Gamma,\Theta}, \rho] + \mathcal{D}_{u,\Gamma,\Theta}[\rho], \quad (6)$$

where  $\mathcal{D}$  is in standard GKLS form. Here  $\Theta \in \mathcal{P}$  is a parameter encoding long-time-scale structure (e.g. admissible-future specifications); see Section 6. The reachable set from  $\rho$  over time  $T$  under  $\Gamma, \Theta$  is

$$\mathcal{R}_T(\rho \mid \Gamma, \Theta) = \{\Phi_{u,T}^{\Gamma,\Theta}(\rho) : u(\cdot) \in \mathcal{U}\}, \quad (7)$$

where  $\Phi_{u,T}^{\Gamma,\Theta}$  denotes the evolution map.

### 3 Three Time-Scales and the Hierarchical Setting

We posit three intrinsic time-scales,

$$\tau_{\text{local}} \ll \tau_{\text{order}} \ll \tau_{\text{time}}, \quad (8)$$

giving rise to two small parameters,

$$\varepsilon_l := \tau_{\text{local}}/\tau_{\text{order}}, \quad \varepsilon_o := \tau_{\text{order}}/\tau_{\text{time}}. \quad (9)$$

The three rationalities act on objects of corresponding time-scales:

| Rationality      | Object                                                 | Time-scale            |
|------------------|--------------------------------------------------------|-----------------------|
| Local            | state $\rho \in \mathcal{D}(\mathcal{H}_{\text{sys}})$ | $\tau_{\text{local}}$ |
| Order-preserving | coupling $\Gamma \in \mathcal{C}_N$                    | $\tau_{\text{order}}$ |
| Time-preserving  | long-time parameter $\Theta \in \mathcal{P}$           | $\tau_{\text{time}}$  |

The slow long-time parameter  $\Theta$  encodes the admissible-future specification  $\mathcal{F}_{\text{adm}} = \mathcal{F}_{\text{adm}}(\Theta)$  together with any other parameters that drift on the longest time-scale. We treat  $\Theta \in \mathcal{P} \subseteq \mathbb{R}^p$  as a finite-dimensional smooth manifold for simplicity.

#### 3.1 Why three layers

The three-fold separation is not arbitrary. Two-layer (state vs. parameter) and four-layer (state, fast coupling, slow coupling, parameter) decompositions are mathematically possible, but the three-layer split corresponds to a structural distinction in the type of object being optimized:

- The state  $\rho \in \mathcal{D}(\mathcal{H}_{\text{sys}})$  is a vector in a real-affine space; its dynamics are linear (Lindbladian) on the tangent space.
- The coupling  $\Gamma \in \mathcal{C}_N$  is a network structure; its dynamics involve combinatorial constraints (Cheeger conductance, subset domination) that are not captured by tangent-space linearity.
- The parameter  $\Theta \in \mathcal{P}$  is a specification of the admissible future; its dynamics encode external boundary conditions on the system's trajectory.

These three categories of object—vector state, network structure, boundary specification—are categorically distinct in the sense that no smooth equivariant map between two of them is in general available. The three-layer decomposition is therefore canonical, not chosen.

Remark 3.1. A four-layer decomposition splitting  $\Gamma$  into “fast modes” and “slow modes” is sometimes useful (e.g. when the coupling itself has multi-scale structure), but the additional split is on the same categorical level, namely network structure, and so is a refinement rather than a new layer in the present sense.

## 4 Local Rationality

### 4.1 Definition

For each node  $i$ , fix a smooth utility  $U_i : \mathcal{D}(\mathcal{H}_{\text{sys}}) \times \mathcal{U} \rightarrow \mathbb{R}$  depending only on the reduced state  $\rho_i$  and the local component  $u_i$  of the control. The local rationality cost is

$$J_{\text{local}}(\rho, u) := - \sum_{i=1}^N a_i U_i(\rho_i, u_i), \quad a_i > 0. \quad (10)$$

## 4.2 Externality

Local optimization can damage inter-node coupling. We define the externality cost as the rate of coupling decay,

$$E_{\text{local}}(\dot{\Gamma}) := \sum_{i \neq j} [-\dot{\gamma}_{ij}]_+^2, \quad [x]_+ := \max(x, 0). \quad (11)$$

The corrected local cost is

$$J_{\text{local}}^*(\rho, u; \dot{\Gamma}) := J_{\text{local}}(\rho, u) + \alpha E_{\text{local}}(\dot{\Gamma}), \quad \alpha \geq 0. \quad (12)$$

We refer to  $\alpha = 0$  as naive local rationality and  $\alpha > 0$  as externality-corrected local rationality.

## 4.3 Decoupling of fast state dynamics from coupling rate

A subtle point is that  $E_{\text{local}}$  depends on  $\dot{\Gamma}$ , which on the fast time-scale is a free variable from the perspective of the state  $\rho$ . The combined system is consistent because, on the fast time-scale of  $\rho$ , both  $\Gamma$  and  $\dot{\Gamma}$  are frozen at their layer-instantaneous values; on the intermediate time-scale of  $\Gamma$ , the state  $\rho$  is approximately constant and  $\dot{\Gamma}$  is determined by the gradient flow of  $J_{\text{order}}$ .

**Lemma 4.1** (Externality term respects time-scale separation). For the three-layer system (28)–(30), the limiting layer dynamics for  $\rho$  in the limit  $\varepsilon_l \rightarrow 0$  (with  $\varepsilon_o > 0$  fixed) is

$$\dot{\rho} = -\nabla_{\rho} J_{\text{local}}(\rho, u) + O(\varepsilon_l),$$

i.e. the externality term  $\alpha E_{\text{local}}(\dot{\Gamma})$  does not contribute to the leading-order fast dynamics of  $\rho$ .

Proof. By definition,  $E_{\text{local}}$  is a function of  $\dot{\Gamma}$  and not of  $\rho$ , hence  $\nabla_{\rho} E_{\text{local}}(\dot{\Gamma}) = 0$  identically. Therefore  $\nabla_{\rho} J_{\text{local}}^* = \nabla_{\rho} J_{\text{local}}$  exactly. The  $O(\varepsilon_l)$  remainder comes from the standard singular-perturbation correction.  $\square$

The lemma shows that adding the externality term to the local cost does not interfere with the fast-state dynamics; it acts only at the intermediate level, through the coupling between the state-time-scale value of  $\dot{\Gamma}_{\text{eq}}$  and the fast dynamics. This is precisely what is needed for the three-layer architecture to be self-consistent.

## 5 Order-Preserving Rationality

This is the central technical section. Following the discussion in [2], we separate structural admissibility (intrinsic to the graph  $\Gamma$ ) from dynamic admissibility (state- and parameter-dependent), and we introduce an admissible-flow cone  $\mathcal{K}_{\text{adm}}$  in addition to the admissible-state set.

### 5.1 Structural admissibility

**Definition 5.1** (Structural admissibility). Fix constants  $d_{\min} > 0$  and  $\phi_{\min} \in (0, 1)$ . The structural admissibility set is

$$\mathcal{O}_{\text{adm}}^{\text{str}} := \left\{ \Gamma \in \mathcal{C}_N : \begin{array}{ll} d_i(\Gamma) \geq d_{\min} & \forall i \in V, \\ \Phi_{\Gamma}(S) \geq \phi_{\min} & \forall S \text{ with } \emptyset \subsetneq S \subsetneq V \end{array} \right\}. \quad (13)$$

The first condition (A1) prevents node-level isolation. The second (A2) prevents subsystem closure, that is, the formation of a coalition  $S$  with weak external connectivity. By the Cheeger inequality, A2 is equivalent to a spectral gap condition on the normalized graph Laplacian,  $\lambda_2 \geq \phi_{\min}^2/2$ , and is thus numerically verifiable.

**Lemma 5.2** (Closedness and continuity). The set  $\mathcal{O}_{\text{adm}}^{\text{str}}$  is closed in the standard topology of  $\mathbb{R}^{N \times N}$ . The map  $\Gamma \mapsto (d_i(\Gamma), \Phi_\Gamma(S))$  is continuous on the open set  $\{\Gamma : \text{vol}_\Gamma(S), \text{vol}_\Gamma(S^c) > 0\}$ , which contains the interior of  $\mathcal{O}_{\text{adm}}^{\text{str}}$ .

*Proof.* Each  $d_i(\Gamma)$  is linear in  $\Gamma$ , hence continuous. The cut  $\text{cut}_\Gamma(S, S^c)$  is also linear. The minimum of two linear functionals is continuous and bounded away from zero on the interior. Hence  $\Phi_\Gamma(S)$  is continuous on the interior. The complement of  $\mathcal{O}_{\text{adm}}^{\text{str}}$  is the union of finitely many open conditions  $\{d_i(\Gamma) < d_{\min}\} \cup \bigcup_S \{\Phi_\Gamma(S) < \phi_{\min}\}$ , hence  $\mathcal{O}_{\text{adm}}^{\text{str}}$  is closed.  $\square$

**Remark 5.3.**  $\mathcal{O}_{\text{adm}}^{\text{str}}$  is closed but generally not compact: there is no upper bound on the entries  $\gamma_{ij} \geq 0$ . When compactness is required (e.g. for application of Tikhonov-type theorems on a bounded domain), one may impose an additional norm bound  $\sum_{i < j} \gamma_{ij} \leq G_{\max}$  or component-wise cap  $\gamma_{ij} \leq \gamma_{\max}$ , restricting the analysis to a compact slice of  $\mathcal{O}_{\text{adm}}^{\text{str}}$ . We assume implicitly throughout the rest of the paper that the trajectories of interest remain in such a compact slice; this is automatic in the numerical experiment of Section 9.

## 5.2 Dynamic admissibility

**Definition 5.4** (Dynamic admissibility). Let  $T > 0$  be a fixed planning horizon and  $P : \mathcal{D}(\mathcal{H}_{\text{sys}}) \times \mathcal{C}_N \rightarrow \mathbb{R}$  a smooth constraint function. The dynamic admissibility set, parameterized by  $(\rho, \Theta)$ , is

$$\mathcal{O}_{\text{adm}}^{\text{dyn}}(\rho, \Theta) := \{\Gamma \in \mathcal{C}_N : \mathcal{R}_T(\rho \mid \Gamma, \Theta) \cap \mathcal{F}_{\text{adm}}(\Theta) \neq \emptyset, \quad P(\rho, \Gamma) \leq 0\}. \quad (14)$$

The first condition (A3) ensures that under coupling  $\Gamma$ , some admissible future is reachable. The second (A4) is the constraint analogue of the PFH (Photon Future Health) ethical constraint of [1] and is treated abstractly here.

## 5.3 Combined admissibility

**Definition 5.5** (Admissible-order set).

$$\mathcal{O}_{\text{adm}}(\rho, \Theta) := \mathcal{O}_{\text{adm}}^{\text{str}} \cap \mathcal{O}_{\text{adm}}^{\text{dyn}}(\rho, \Theta). \quad (15)$$

We assume throughout the paper that the admissible-order set is non-empty and has non-empty interior:

**Assumption 5.6** (Strict feasibility). For all  $(\rho, \Theta)$  in the relevant compact set,  $\text{int } \mathcal{O}_{\text{adm}}(\rho, \Theta) \neq \emptyset$ .

## 5.4 Admissible-flow cone

The state set  $\mathcal{O}_{\text{adm}}$  alone is insufficient: a  $\Gamma \in \mathcal{O}_{\text{adm}}$  may be evolving in a direction  $\dot{\Gamma}$  that, while not yet leaving  $\mathcal{O}_{\text{adm}}$ , indicates incipient subsystem closure. We introduce a flow-level condition.

**Definition 5.7** (Domination indicator and admissible-flow cone). For  $S \subsetneq V$  with  $S, S^c \neq \emptyset$ , let

$$A_S(\dot{\Gamma}) = \frac{1}{|S|(|S|-1)} \sum_{\substack{i,j \in S \\ i \neq j}} \dot{\gamma}_{ij}, \quad (16)$$

$$B_S(\dot{\Gamma}) = \frac{1}{|S||S^c|} \sum_{\substack{i \in S \\ j \in S^c}} \dot{\gamma}_{ij}. \quad (17)$$

For a fixed family of subsets  $\mathcal{S}_* \subseteq 2^V$  and weights  $\alpha_S, \beta_S > 0$ , define the convex domination indicator

$$D_{\text{dom}}(\dot{\Gamma}) := \sum_{S \in \mathcal{S}_*} (\alpha_S [A_S(\dot{\Gamma})]_+ + \beta_S [-B_S(\dot{\Gamma})]_+). \quad (18)$$

For  $\theta \geq 0$ , the admissible-flow cone is

$$\mathcal{K}_{\text{adm}} := \{\dot{\Gamma} \in \mathbb{R}_{\text{sym},0}^{N \times N} : D_{\text{dom}}(\dot{\Gamma}) \leq \theta\}, \quad (19)$$

where  $\mathbb{R}_{\text{sym},0}^{N \times N}$  denotes symmetric zero-diagonal matrices.

Remark 5.8. The original (non-convex) indicator

$$D_{\text{dom}}(\dot{\Gamma}) = \sup_{S \subsetneq V} [A_S(\dot{\Gamma})]_+ \cdot [-B_S(\dot{\Gamma})]_+$$

captures the supremum-of-products form most directly tied to the intuitive picture of subsystem closure. The convex relaxation  $D_{\text{dom}}$  above is computationally tractable and admits a well-defined orthogonal projection  $\Pi_{\mathcal{K}_{\text{adm}}}$ . In Paper 2 we will return to the non-convex version.

**Lemma 5.9** (Properties of  $\mathcal{K}_{\text{adm}}$ ).  $\mathcal{K}_{\text{adm}}$  is a closed convex cone (when  $\theta = 0$ ) or a closed convex set (when  $\theta > 0$ ). The map  $\dot{\Gamma} \mapsto D_{\text{dom}}(\dot{\Gamma})$  is convex, continuous, and piecewise linear. The metric projection  $\Pi_{\mathcal{K}_{\text{adm}}}$  onto  $\mathcal{K}_{\text{adm}}$  is well-defined and Lipschitz with constant 1.

Proof.  $A_S$  and  $B_S$  are linear functionals, hence  $[A_S]_+$  and  $[-B_S]_+$  are convex piecewise linear functions of  $\dot{\Gamma}$ , as is their non-negative combination  $D_{\text{dom}}$ . Convex sublevel sets are closed and convex; for  $\theta = 0$ , the set is also a cone since  $D_{\text{dom}}$  is positively homogeneous. The projection onto a closed convex set in finite dimensions is well-defined and 1-Lipschitz.  $\square$

## 5.5 Order-preserving cost

**Definition 5.10** (Order-preserving cost). The order-preserving cost is

$$J_{\text{order}}(\rho, \Gamma, \dot{\Gamma}; \Theta) := d(\Gamma, \mathcal{O}_{\text{adm}}(\rho, \Theta))^2 + \chi D_{\text{dom}}(\dot{\Gamma}) + \beta \sum_{i \neq j} w_{ij} (I(\rho_{ij}) - I_{ij}^*)^2, \quad (20)$$

where  $d(\cdot, \mathcal{O}_{\text{adm}})$  is the Euclidean distance to  $\mathcal{O}_{\text{adm}}$  in  $\mathcal{C}_N$ ,  $I(\rho_{ij}) = S(\rho_i) + S(\rho_j) - S(\rho_{ij})$  is the mutual information of the bipartite reduced state,  $I_{ij}^*$  are reference values, and  $\chi, \beta, w_{ij} > 0$ .

The three terms penalize: (i) deviation of the present coupling from admissibility, (ii) closure-prone evolution direction, (iii) deviation of bipartite mutual informations from reference values.



## 5.6 Geometric properties of $\mathcal{K}_{\text{adm}}$

The cone  $\mathcal{K}_{\text{adm}}$  has a useful explicit description and a precise relationship to the original non-convex condition.

**Proposition 5.11** (Explicit description of  $\mathcal{K}_{\text{adm}}$  for  $|\mathcal{S}_*| = 1$ ). If  $\mathcal{S}_* = \{S\}$  is a single subset,  $\theta = 0$ , and  $\alpha_S = \beta_S = 1$ , then  $D_{\text{dom}}(\dot{\Gamma}) = [A_S]_+ + [-B_S]_+$ , and

$$\mathcal{K}_{\text{adm}} = \{\dot{\Gamma} : A_S(\dot{\Gamma}) \leq 0 \text{ and } B_S(\dot{\Gamma}) \geq 0\}.$$

That is,  $\mathcal{K}_{\text{adm}}$  excludes any  $\dot{\Gamma}$  either strengthening internal couplings ( $A_S > 0$ ) or weakening external couplings ( $B_S < 0$ ).

Proof.  $D_{\text{dom}}^{\sim} = 0$  requires both non-negative summands  $[A_S]_+$  and  $[-B_S]_+$  to vanish, which is equivalent to  $A_S \leq 0$  and  $B_S \geq 0$  simultaneously.  $\square$

## 5.7 Conservativeness of the convex relaxation

This is a crucial point that should be made transparent. Let

$$D_{\text{dom}}(\dot{\Gamma}) := \sup_{S \in \mathcal{S}_*} [A_S(\dot{\Gamma})]_+ \cdot [-B_S(\dot{\Gamma})]_+ \quad (21)$$

denote the original non-convex domination indicator (whose vanishing was the heuristically correct condition motivating the design of  $\mathcal{K}_{\text{adm}}$ ). Then  $D_{\text{dom}}(\dot{\Gamma}) = 0$  if and only if for every  $S$ ,  $A_S \leq 0$  or  $B_S \geq 0$  (the negation of simultaneous internal strengthening and external weakening at  $S$ ). Compare this with the convex condition  $D_{\text{dom}}^{\sim}(\dot{\Gamma}) \leq 0$ , which by Proposition 5.11 (and its multi-subset extension) requires for every  $S$  that  $A_S \leq 0$  and  $B_S \geq 0$ .

The disjunction in the original condition has been replaced by a conjunction in the convex relaxation. We make this explicit:

**Proposition 5.12** (Conservativeness of the convex relaxation). Let  $\mathcal{K}_{\text{adm}}^{\text{nc}} := \{\dot{\Gamma} : D_{\text{dom}}(\dot{\Gamma}) = 0\}$  denote the original non-convex admissible-flow set, and  $\mathcal{K}_{\text{adm}} := \{\dot{\Gamma} : D_{\text{dom}}^{\sim}(\dot{\Gamma}) \leq 0\}$  the convex relaxation (with  $\theta = 0$ ,  $\mathcal{S}_*$  arbitrary, and  $\alpha_S, \beta_S > 0$ ). Then

$$\mathcal{K}_{\text{adm}} \subseteq \mathcal{K}_{\text{adm}}^{\text{nc}}, \quad (22)$$

and the inclusion is generally strict.

Proof. If  $D_{\text{dom}}^{\sim}(\dot{\Gamma}) \leq 0$ , then since each summand  $\alpha_S[A_S]_+ + \beta_S[-B_S]_+$  is non-negative, every summand vanishes. Thus for every  $S$ ,  $[A_S]_+ = 0$  and  $[-B_S]_+ = 0$ , i.e.  $A_S \leq 0$  and  $B_S \geq 0$ . In particular,  $A_S \leq 0$  or  $B_S \geq 0$  for every  $S$ , so  $D_{\text{dom}}(\dot{\Gamma}) = 0$ .

For the strictness of the inclusion: take  $|V| = 3$ ,  $S = \{1, 2\}$  in  $\mathcal{S}_*$ , and  $\dot{\Gamma}$  with  $\dot{\gamma}_{12} = +1$ ,  $\dot{\gamma}_{13} = +1$ ,  $\dot{\gamma}_{23} = 0$ . Then  $A_S = 1 > 0$  and  $B_S = 1/2 > 0$ , so  $[A_S]_+ \cdot [-B_S]_+ = 1 \cdot 0 = 0$  and  $D_{\text{dom}}^{\sim}(\dot{\Gamma}) = 0$  (the original condition holds, since  $B_S \geq 0$ ). However  $\alpha_S[A_S]_+ + \beta_S[-B_S]_+ = \alpha_S > 0$ , so  $D_{\text{dom}}(\dot{\Gamma}) > 0$  and  $\dot{\Gamma} \notin \mathcal{K}_{\text{adm}}$ .  $\square$

**Remark 5.13** (Physical meaning of the conservativeness). The convex relaxation has a definite physical content beyond mathematical convenience. The original (non-convex) condition prohibits a node coalition  $S$  from simultaneously strengthening its internal couplings while severing its external couplings—the textbook scenario of subsystem closure. The convex relaxation strengthens this to: a coalition  $S$  may neither (independently) strengthen its internal couplings nor (independently) weaken its external ones. Read in conjunction with the environment-shared coupling

structure of [2], this stronger condition expresses the impossibility of a coalition unilaterally consolidating itself “without paying its share” of the shared environment. The conservativeness is therefore not a purely numerical artefact: it encodes a no-free-rider principle on coupling change. The cost of this principled strengthening is a more conservative cone; the benefit is that the projection  $\Pi_{\mathcal{K}_{\text{adm}}}$  becomes a tractable convex projection rather than a non-convex selector.

**Remark 5.14.** The original non-convex  $D_{\text{dom}}$  in (21) can be recovered as a refinement (Paper 2). The  $\theta > 0$  slack in the definition of  $\mathcal{K}_{\text{adm}}$  partially relaxes the strict inclusion of Proposition 5.12, but the disjunctive flexibility of the original condition is not fully recovered for any finite  $\theta$ .

**Proposition 5.15** (Compatibility with the structural cone). If  $\Gamma \in \text{int } \mathcal{O}_{\text{adm}}^{\text{str}}$  and  $\dot{\Gamma} \in \mathcal{K}_{\text{adm}}$  with  $\theta = 0$  and  $\mathcal{S}_*$  containing all proper subsets, then for sufficiently small  $h > 0$ ,

$$\Gamma + h \dot{\Gamma} \in \mathcal{O}_{\text{adm}}^{\text{str}}.$$

*Proof.* A2 (Cheeger conductance) is preserved infinitesimally if and only if for all proper subsets  $S$ ,

$$\left. \frac{d}{dh} \Phi_{\Gamma+h\dot{\Gamma}}(S) \right|_{h=0} \geq 0.$$

A direct computation gives

$$\frac{d}{dh} \Phi(S) = \frac{|S||S^c|B_S(\dot{\Gamma})}{\min\{\text{vol}(S), \text{vol}(S^c)\}} - \frac{\text{cut}(S, S^c)}{[\min\{\text{vol}(S), \text{vol}(S^c)\}]^2} \cdot \frac{d}{dh} \min\{\text{vol}(S), \text{vol}(S^c)\}.$$

For  $\dot{\Gamma} \in \mathcal{K}_{\text{adm}}$ ,  $B_S \geq 0$  for every  $S$ , so the leading term is non-negative, and  $A_S \leq 0$  controls the volume derivative term. Combined,  $d\Phi/dh \geq 0$ . A1 (degree lower bound) is similarly preserved by  $A_S \leq 0$  for the corresponding singletons.  $\square$

The propositions show that the convex relaxation  $D_{\text{dom}}^{\sim}$  inherits the desired structural meaning: a  $\dot{\Gamma}$  in  $\mathcal{K}_{\text{adm}}$  is geometrically a direction of motion that does not push  $\Gamma$  out of the structural admissible region.

## 6 Time-Preserving Rationality

### 6.1 Reachable and admissible futures

The reachable set  $\mathcal{R}_T(\rho \mid \Gamma, \Theta)$  was introduced in Section 2. The admissible-future set  $\mathcal{F}_{\text{adm}}(\Theta) \subseteq \mathcal{D}(\mathcal{H}_{\text{sys}})$  is fixed externally and parameterized by  $\Theta \in \mathcal{P}$ . We assume:

**Assumption 6.1.**  $\mathcal{F}_{\text{adm}}(\Theta)$  is closed and depends continuously on  $\Theta$  in Hausdorff distance, and  $\Theta \mapsto \mathcal{F}_{\text{adm}}(\Theta)$  is locally Lipschitz.

### 6.2 Future volume

**Definition 6.2** (Future volume). On the affine hyperplane of trace-one Hermitian operators in  $\mathcal{B}(\mathcal{H}_{\text{sys}})$ , equipped with Hilbert–Schmidt measure  $\mu_{\text{HS}}$ , define

$$V_{\text{future}}(\rho; \Gamma, \Theta) := \mu_{\text{HS}}(\mathcal{R}_T(\rho \mid \Gamma, \Theta) \cap \mathcal{F}_{\text{adm}}(\Theta)). \quad (23)$$

In the finite-dimensional setting this is well-defined. Because the reachable set may be lower-dimensional,  $V_{\text{future}}$  may equal zero generically; we use  $V_{\text{future}} + \varepsilon$  in cost expressions to avoid singularity, and replace  $\mu_{\text{HS}}$  by  $\mu_{\text{HS}}$  on the affine span of  $\mathcal{R}_T$  when generic vanishing is the issue.

**Lemma 6.3** (Upper semicontinuity of  $V_{\text{future}}$ ). Under Assumption 6.1, the function  $(\rho, \Gamma, \Theta) \mapsto V_{\text{future}}(\rho; \Gamma, \Theta)$  is upper semicontinuous on its domain.

Proof. The reachable set  $\mathcal{R}_T(\rho \mid \Gamma, \Theta)$  is the image of a compact control set  $\mathcal{U}$  under the continuous map  $u \mapsto \Phi_{u,T}^{\Gamma, \Theta}(\rho)$ , hence compact and continuously varying in Hausdorff distance with  $(\rho, \Gamma, \Theta)$ . Together with continuous Hausdorff variation of  $\mathcal{F}_{\text{adm}}(\Theta)$  from Assumption 6.1, the intersection  $\mathcal{R}_T \cap \mathcal{F}_{\text{adm}}(\Theta)$  is upper semicontinuous in Hausdorff distance. Hilbert–Schmidt measure of an upper-semicontinuously varying compact family is upper semicontinuous.  $\square$

Remark 6.4.  $V_{\text{future}}$  is generally not lower semicontinuous: a small perturbation can shrink the reachable set to a measure-zero subset (e.g. when the new dynamics is unitary while the original was not). The use of  $\log(V_{\text{future}} + \varepsilon)$  in  $J_{\text{time}}$  ensures gradient-based dynamics still behave well: although  $\log(V_{\text{future}} + \varepsilon)$  may have discontinuities of the right sign, gradient descent will not be attracted toward them.

### 6.3 Irreversibility

**Definition 6.5** (Irreversibility cost). Let  $\rho(\cdot)$  be a state trajectory with  $\rho(0) = \rho_0$  on  $[0, t]$ . Define the unitary orbit of  $\rho_0$ ,

$$\mathcal{U}(\rho_0) := \{U\rho_0U^\dagger : U \in \mathcal{U}(\mathcal{H}_{\text{sys}})\}. \quad (24)$$

The trace-distance to the unitary orbit is

$$d_{\mathcal{U}}(\rho(s), \rho_0) := \inf_{U \in \mathcal{U}(\mathcal{H}_{\text{sys}})} \|\rho(s) - U\rho_0U^\dagger\|_1. \quad (25)$$

The irreversibility cost up to time  $t$  is

$$I_{\text{irr}}(t) := \int_0^t d_{\mathcal{U}}(\rho(s), \rho_0)^2 ds. \quad (26)$$

**Lemma 6.6** (Properties of  $I_{\text{irr}}$ ).  $I_{\text{irr}}(t) \geq 0$  for all trajectories. Moreover  $I_{\text{irr}}(t) = 0$  if and only if  $\rho(\cdot)$  is unitarily equivalent to  $\rho_0$  on  $[0, t]$ .

Proof. Non-negativity is immediate. Vanishing implies  $d_{\mathcal{U}}(\rho(s), \rho_0) = 0$  for almost every  $s$ . Since the unitary orbit is closed (image of a compact group action) and  $\rho$  is continuous, this means  $\rho(s) \in \mathcal{U}(\rho_0)$  for all  $s$ , i.e.  $\rho(s) = U(s)\rho_0U(s)^\dagger$  for some measurable  $U(\cdot)$ .  $\square$

### 6.4 Time-preserving cost

**Definition 6.7** (Time-preserving cost).

$$J_{\text{time}}(\rho, \Gamma, \Theta) := -\log(V_{\text{future}}(\rho; \Gamma, \Theta) + \varepsilon) + \kappa I_{\text{irr}} + \eta E_{\text{present}} + \zeta E_{\text{model}}, \quad (27)$$

where  $E_{\text{present}}(\rho, u)$  measures present-cost on nodes (defined in Appendix A) and  $E_{\text{model}}$  measures predictive uncertainty across the family of forward models. The coefficients  $\kappa, \eta, \zeta > 0$ .

## 7 Hierarchical Optimization and Singular Perturbation

### 7.1 Three-layer dynamics

The combined dynamics, on the long-time variable  $s = t/\tau_{\text{time}}$ , take the form

$$\varepsilon_l \varepsilon_o \frac{d\rho}{ds} = -\nabla_{\rho} J_{\text{local}}^*(\rho, u; \dot{\Gamma}), \quad (28)$$

$$\varepsilon_o \frac{d\Gamma}{ds} = \Pi_{\mathcal{K}_{\text{adm}}} [-\nabla_{\Gamma} J_{\text{order}}(\rho, \Gamma, \dot{\Gamma}; \Theta)], \quad (29)$$

$$\frac{d\Theta}{ds} = -\nabla_{\Theta} J_{\text{time}}(\rho, \Gamma, \Theta). \quad (30)$$

Here  $\nabla_{\rho}$  denotes the gradient with respect to the Hilbert–Schmidt structure on the affine space of Hermitian operators with trace one.

### 7.2 Two-layer reduction

The two-layer reduction is obtained by formally setting  $\varepsilon_o = 0$  in (29), replacing  $\Gamma$  by its instantaneous equilibrium

$$\Gamma_{\text{eq}}(\rho, \Theta) := \arg \min_{\Gamma \in \mathcal{O}_{\text{adm}}(\rho, \Theta)} J_{\text{order}}(\rho, \Gamma, 0; \Theta). \quad (31)$$

**Assumption 7.1** (Well-defined slow manifold). The minimizer in (31) exists, is unique, lies in the interior of  $\mathcal{O}_{\text{adm}}(\rho, \Theta)$ , and depends  $C^2$  on  $(\rho, \Theta)$  on the relevant compact domain. Moreover the Hessian  $\nabla_{\Gamma}^2 J_{\text{order}}$  is uniformly positive definite at  $\Gamma = \Gamma_{\text{eq}}$ .

The two-layer dynamics are

$$\varepsilon_l \frac{d\rho}{ds} = -\nabla_{\rho} J_{\text{local}}^*(\rho, u; \dot{\Gamma}_{\text{eq}}), \quad (32)$$

$$\frac{d\Theta}{ds} = -\nabla_{\Theta} J_{\text{time}}(\rho, \Gamma_{\text{eq}}(\rho, \Theta), \Theta). \quad (33)$$

### 7.3 Tikhonov-type slow-manifold lemma

**Lemma 7.2** (Slow-manifold convergence). Suppose Assumptions 5.6, 6.1, 7.1 hold. Let  $\rho^{(\varepsilon_o)}, \Gamma^{(\varepsilon_o)}, \Theta^{(\varepsilon_o)}$  be the solution to (28)–(30) on a fixed compact interval  $[0, T]$ , with initial conditions  $(\rho_0, \Gamma_0, \Theta_0)$  such that  $\Gamma_0 = \Gamma_{\text{eq}}(\rho_0, \Theta_0)$ . Then

$$\sup_{s \in [0, T]} \|\Gamma^{(\varepsilon_o)}(s) - \Gamma_{\text{eq}}(\rho^{(\varepsilon_o)}(s), \Theta^{(\varepsilon_o)}(s))\| \leq K \varepsilon_o \quad (34)$$

for some constant  $K = K(T)$  independent of  $\varepsilon_o$ .

*Proof.* We give a self-contained proof along the lines of Fenichel’s geometric singular perturbation theory [3, 4], adapted to our finite-dimensional setting.

**Step 1: Fast variable.** On the long-time variable  $s = t/\tau_{\text{time}}$ , equation (29) reads

$$\frac{d\Gamma}{ds} = \frac{1}{\varepsilon_o} \Pi_{\mathcal{K}_{\text{adm}}} [-\nabla_{\Gamma} J_{\text{order}}(\rho, \Gamma; \Theta)].$$

Setting  $\sigma = s/\varepsilon_o$  (the fast time-scale), we obtain the layer system

$$\frac{d\Gamma}{d\sigma} = \Pi_{\mathcal{K}_{\text{adm}}} [-\nabla_{\Gamma} J_{\text{order}}(\rho, \Gamma; \Theta)], \quad (35)$$

with  $(\rho, \Theta)$  frozen at their initial values.

**Step 2: Layer dynamics is a gradient flow on a strictly convex landscape.** By Assumption 7.1, the function  $\Gamma \mapsto J_{\text{order}}(\rho, \Gamma; \Theta)$  has a unique minimizer  $\Gamma_{\text{eq}}(\rho, \Theta)$  in the interior of  $\mathcal{O}_{\text{adm}}(\rho, \Theta)$ , with Hessian uniformly positive definite,

$$\nabla_{\Gamma}^2 J_{\text{order}} \succeq m I, \quad m > 0.$$

The projection  $\Pi_{\mathcal{K}_{\text{adm}}}$  onto the closed convex cone  $\mathcal{K}_{\text{adm}}$  is 1-Lipschitz. Note that since  $\mathcal{K}_{\text{adm}}$  is a closed convex cone (and not a linear subspace),  $\Pi_{\mathcal{K}_{\text{adm}}}[-\nabla J]$  does not in general equal  $-\Pi_{\mathcal{K}_{\text{adm}}}[\nabla J]$ ; we therefore project the descent direction  $-\nabla_{\Gamma} J_{\text{order}}$  rather than the gradient itself, so that the projected vector field is itself a valid element of  $\mathcal{K}_{\text{adm}}$ . The flow (35) therefore satisfies

$$\frac{d}{d\sigma} \frac{1}{2} \|\Gamma - \Gamma_{\text{eq}}\|^2 = \langle \Gamma - \Gamma_{\text{eq}}, \Pi_{\mathcal{K}_{\text{adm}}}[-\nabla_{\Gamma} J_{\text{order}}] \rangle \leq -m \|\Gamma - \Gamma_{\text{eq}}\|^2 + r(\Gamma),$$

where  $r$  collects projection error terms vanishing at  $\Gamma_{\text{eq}}$ . Since  $\nabla_{\Gamma} J_{\text{order}}(\rho, \Gamma_{\text{eq}}; \Theta) = 0$  and  $\Gamma_{\text{eq}} \in \text{int } \mathcal{K}_{\text{adm}}$  (a consequence of strict feasibility), there is a neighborhood in which  $\Pi_{\mathcal{K}_{\text{adm}}}[-\nabla_{\Gamma} J_{\text{order}}] = -\nabla_{\Gamma} J_{\text{order}}$  and  $r \equiv 0$ . Hence on the  $\sigma$ -scale,

$$\|\Gamma(\sigma) - \Gamma_{\text{eq}}\| \leq e^{-m\sigma} \|\Gamma(0) - \Gamma_{\text{eq}}\|. \quad (36)$$

This is exponential attraction toward the slow manifold on the fast time-scale, with rate  $\lambda := m$ .

**Step 3: Slow manifold persistence.** Define the slow manifold

$$M := \{(\rho, \Gamma, \Theta) : \Gamma = \Gamma_{\text{eq}}(\rho, \Theta)\}.$$

By the implicit function theorem (using the non-degeneracy of  $\nabla_{\Gamma}^2 J_{\text{order}}$ ),  $M$  is a smooth submanifold of codimension equal to  $\dim \Gamma$ . Fenichel's theorem [3] guarantees that for small enough  $\varepsilon_o > 0$ , there exists an invariant manifold  $M_{\varepsilon_o}$  which is  $C^k$ -close to  $M$  uniformly on compacts, with

$$\text{dist}_{C^0}(M_{\varepsilon_o}, M) = O(\varepsilon_o).$$

**Step 4: Trajectory tracking.** Initial condition  $\Gamma_0 = \Gamma_{\text{eq}}(\rho_0, \Theta_0)$  lies on  $M$ . The exact trajectory then lies on  $M_{\varepsilon_o}$  for all time. Combining with Step 3,

$$\|\Gamma^{(\varepsilon_o)}(s) - \Gamma_{\text{eq}}(\rho^{(\varepsilon_o)}(s), \Theta^{(\varepsilon_o)}(s))\| \leq K \varepsilon_o, \quad s \in [0, T],$$

for some  $K = K(T)$  independent of  $\varepsilon_o$ . □

## 8 Buffer Stability Theorem

We now state and prove the two main results.

### 8.1 Feasibility sets

For the three-layer system, the feasibility set is

$$\mathcal{I}_3(T) := \{(\rho_0, \Gamma_0, \Theta_0) : \exists u(\cdot) \in \mathcal{U} \text{ s.t. admissibility holds on } [0, T]\}, \quad (37)$$

where “admissibility holds on  $[0, T]$ ” means  $\Gamma(s) \in \mathcal{O}_{\text{adm}}(\rho(s), \Theta(s))$  and  $P(\rho(s), \Gamma(s)) \leq 0$  for all  $s \in [0, T]$  along the trajectory of (28)–(30). For the two-layer system,

$$\mathcal{I}_2(T) := \{(\rho_0, \Theta_0) : \exists u(\cdot) \in \mathcal{U} \text{ s.t. admissibility holds for the trajectory of (32), (33)}\}. \quad (38)$$

Define the slow-manifold lift

$$\text{lift}(\rho_0, \Theta_0) := (\rho_0, \Gamma_{\text{eq}}(\rho_0, \Theta_0), \Theta_0). \quad (39)$$

## 8.2 Strict admissibility and the lift theorem

To formulate the feasibility lift cleanly, we first isolate the strict-admissibility condition as an independent definition.

**Definition 8.1** ( $\delta$ -strictly admissible two-layer trajectory). A two-layer trajectory  $(\rho_2(s), \Theta_2(s))_{s \in [0, T]}$  together with its slow-manifold companion  $\Gamma_{\text{eq}}(\rho_2(s), \Theta_2(s))$  is called  $\delta$ -strictly admissible, for some  $\delta > 0$ , if for every  $s \in [0, T]$ ,

$$\text{dist}(\Gamma_{\text{eq}}(\rho_2(s), \Theta_2(s)), \partial \mathcal{O}_{\text{adm}}(\rho_2(s), \Theta_2(s))) \geq \delta, \quad (40)$$

$$P(\rho_2(s), \Gamma_{\text{eq}}(\rho_2(s), \Theta_2(s))) \leq -\delta. \quad (41)$$

The set of initial conditions  $(\rho_0, \Theta_0)$  admitting such a trajectory is denoted  $\mathcal{I}_2^\delta(T)$ .

The first inequality (40) ensures uniform separation from the structural admissibility boundary, while (41) ensures uniform separation from the constraint boundary  $\{P = 0\}$ . Both are necessary because either kind of grazing can break under singular perturbation.

**Theorem 8.2** (Feasibility Inclusion under Strict Admissibility). Suppose Assumptions 5.6, 6.1, 7.1 hold, and that the slow-manifold tracking estimate of Lemma 7.2 applies. Then there exists  $\varepsilon_0 > 0$  such that for all  $\varepsilon_o \in (0, \varepsilon_0)$ ,

$$\text{lift}(\mathcal{I}_2^\delta(T)) \subseteq \mathcal{I}_3(T). \quad (42)$$

Proof. Fix  $(\rho_0, \Theta_0) \in \mathcal{I}_2^\delta(T)$  with witness control  $u(\cdot)$ , and let  $(\rho_2, \Theta_2)$  be the corresponding  $\delta$ -strictly admissible two-layer trajectory. By Definition 8.1,

$$\Gamma_{\text{eq}}(\rho_2(s), \Theta_2(s)) \in \text{int } \mathcal{O}_{\text{adm}}(\rho_2(s), \Theta_2(s)) \quad \forall s \in [0, T],$$

with margin  $\delta$  in both senses (40)–(41).

Now consider the three-layer system initialized at  $\text{lift}(\rho_0, \Theta_0) = (\rho_0, \Gamma_{\text{eq}}(\rho_0, \Theta_0), \Theta_0)$  and driven by the same control  $u(\cdot)$  on the state. By Lemma 7.2,

$$\|\Gamma_3(s) - \Gamma_{\text{eq}}(\rho_3(s), \Theta_3(s))\| \leq K \varepsilon_o \quad \forall s \in [0, T], \quad (43)$$

for some  $K = K(T)$ .

By continuous dependence of solutions on parameters (standard Gronwall, exploiting smoothness of all generators) and Lipschitz continuity of  $\Gamma_{\text{eq}}$  (Lemma C.1),

$$(\rho_3(s), \Theta_3(s)) \rightarrow (\rho_2(s), \Theta_2(s)) \quad (44)$$

uniformly on  $[0, T]$  as  $\varepsilon_o \rightarrow 0$ . Combining (43), (44), and Lipschitz continuity of  $\Gamma_{\text{eq}}$ ,

$$\|\Gamma_3(s) - \Gamma_{\text{eq}}(\rho_2(s), \Theta_2(s))\| \rightarrow 0 \quad \text{uniformly on } [0, T].$$

Choose  $\varepsilon_0$  small enough that this combined deviation is at most  $\delta/2$  for all  $\varepsilon_o \in (0, \varepsilon_0)$ . Then by (40),  $\Gamma_3(s)$  lies in  $\text{int } \mathcal{O}_{\text{adm}}(\rho_3(s), \Theta_3(s))$  at distance  $\geq \delta/2$  from the boundary (using also Lemma C.2), and by (41) together with smoothness of  $P$  and the same continuity,  $P(\rho_3(s), \Gamma_3(s)) \leq -\delta/2 < 0$ . Both admissibility conditions hold strictly for the three-layer trajectory on  $[0, T]$ , hence  $\text{lift}(\rho_0, \Theta_0) \in \mathcal{I}_3(T)$ .  $\square$

**Remark 8.3.** The strict-margin requirement  $\delta > 0$  is essential. A two-layer trajectory grazing the boundary of admissibility may fail to lift, since the singular perturbation introduces an  $O(\varepsilon_o)$  deviation that can push the trajectory across the boundary. The theorem isolates the lift to the genuinely interior part  $\mathcal{I}_2^\delta(T)$  of  $\mathcal{I}_2(T)$ .

### 8.3 Buffer Stability under Barrier-Regularized Order Flow

The buffer-stability statement makes precise the claim that the three-layer hierarchy strictly dominates the two-layer reduction in terms of constraint violation when initial conditions lie off the slow manifold. As shown in the proof below, the upper bound on the three-layer violation depends not on the Tikhonov tracking estimate alone but on a geometric property of the order-preserving flow with respect to the admissible region  $\{P \leq 0\}$ . We isolate this property as an explicit assumption.

**Assumption 8.4** (Operational barrier property). The order-preserving cost takes the regularized form

$$J_{\text{order}}^B(\rho, \Gamma, \dot{\Gamma}; \Theta) = J_{\text{order}}(\rho, \Gamma, \dot{\Gamma}; \Theta) + b(P(\rho, \Gamma)),$$

where  $b$  belongs to one of the following classes:

- **(B1) True barrier:**  $b : (-\infty, 0) \rightarrow \mathbb{R}$  is  $C^2$ , monotonically increasing, with  $b(r) \rightarrow +\infty$  as  $r \uparrow 0$  from the admissible side. (Example:  $b(r) = -\beta \log(-r)$ .)
- **(B2) Soft penalty:**  $b : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$  is  $C^1$ , with  $b(r) = 0$  for  $r \leq 0$  and  $b(r) > 0$  for  $r > 0$ . (Example:  $b(r) = \beta [r]_+^2$ .)
- **(B3) Collar penalty:**  $b : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$  is  $C^1$ , supported on  $[-\delta_b, +\infty)$  for some  $\delta_b > 0$ , increasing on  $[-\delta_b, +\infty)$ .

The flow generated by  $J_{\text{order}}^B$  then satisfies one of the corresponding operational barrier properties:

- (1) **Invariance form (under (B1) or (B3) with  $\delta_b$  small enough):**  $\Gamma(0) \in \{P(\rho_0, \cdot) \leq 0\} \implies \Gamma(s) \in \{P(\rho(s), \cdot) \leq 0\} \quad \forall s \in [0, T]$ .
- (2) **Exponential re-entry form (under (B2)):** there exist constants  $C_b, \lambda_b > 0$  such that

$$[P(\rho(s), \Gamma(s))]_+ \leq C_b e^{-\lambda_b s / \varepsilon_o} + C_b \varepsilon_o, \quad s \in [0, T].$$

The two operational forms are dual: (1) prevents trajectory entry into  $\{P > 0\}$  entirely (necessary for true barriers), while (2) allows transient entry but bounds its magnitude exponentially in time and linearly in  $\varepsilon_o$  (a soft penalty cannot prevent transient overshoot, but it bounds it operationally). For the proof of Theorem 8.10 below, either form suffices: form (1) makes the integrand of  $V_{3L}^P$  identically zero (modulo state-evolution corrections of size  $O(\varepsilon_o)$ ), while form (2) directly gives the integrand bound.

**Remark 8.5** (Verification in the numerical experiment). The numerical experiment of Section 9 uses a Gaussian-truncated soft repulsion of class (B3) (the barrier strength is finite but acts only in a collar around  $\{P = 0\}$ ). The observed behavior in Figures 1–2 corresponds to the invariance form (1): the three-layer trajectory is deflected entirely around the forbidden region without entry. This is consistent with (B3) with  $\delta_b$  chosen large enough relative to the slow-manifold drift  $K\varepsilon_o$ .

**Definition 8.6** (Off-manifold initialization with feasible endpoints). An initial triple  $(\rho_0, \Gamma_0, \Theta_0)$  is  $\delta_0$ -off-manifold (with feasible endpoints) if

$$\|\Gamma_0 - \Gamma_{\text{eq}}(\rho_0, \Theta_0)\| = \delta_0 > 0, \tag{45}$$

$$P(\rho_0, \Gamma_0) \leq 0, \tag{46}$$

$$P(\rho_0, \Gamma_{\text{eq}}(\rho_0, \Theta_0)) \leq 0, \tag{47}$$

and the line segment  $\{(1-t)\Gamma_0 + t\Gamma_{\text{eq}}(\rho_0, \Theta_0) : t \in [0, 1]\}$  has non-empty intersection with  $\{\Gamma : P(\rho_0, \Gamma) > 0\}$ .

Remark 8.7 (Geometry of the off-manifold initialization). Conditions (46)–(47) require both endpoints—the initial coupling and the slow-manifold equilibrium—to be admissible. The crossing condition (line segment meets  $\{P > 0\}$ ) means that the admissible region is non-convex along the path between them: the straight line passes through a forbidden island. This is the only setting in which the buffer principle has nontrivial content; if the line segment stayed in  $\{P \leq 0\}$ , the two-layer jump would also stay admissible and the comparison would be trivial.

**Definition 8.8** (Two-layer projection trajectory). For an off-manifold initialization, the natural two-layer interpretation is that  $\Gamma$  jumps instantaneously from  $\Gamma_0$  to  $\Gamma_{\text{eq}}(\rho_0, \Theta_0)$ . We model this as a parameterized family with infinitesimal jump time  $\varepsilon_o^*$ :

$$\Gamma_2^{\text{jump}}(s) = \begin{cases} (1 - s/\varepsilon_o^*)\Gamma_0 + (s/\varepsilon_o^*)\Gamma_{\text{eq}}(\rho_0, \Theta_0), & 0 \leq s \leq \varepsilon_o^*, \\ \Gamma_{\text{eq}}(\rho_2(s), \Theta_2(s)), & s > \varepsilon_o^*. \end{cases} \quad (48)$$

The two-layer system with  $\varepsilon_o^* \rightarrow 0$  produces an instantaneous jump.

**Definition 8.9** (PFH-violation integral).

$$V_{[0,T]}^P(\rho(\cdot), \Gamma(\cdot)) := \int_0^T [P(\rho(s), \Gamma(s))]_+ ds. \quad (49)$$

**Theorem 8.10** (Buffer Stability under Barrier-Regularized Order Flow). Suppose Assumptions 5.6, 6.1, 7.1, and 8.4 hold, and additionally:

- (i)  $P : \mathcal{D}(\mathcal{H}_{\text{sys}}) \times \mathcal{C}_N \rightarrow \mathbb{R}$  is  $C^1$  with bounded gradient  $\|\nabla_{\Gamma} P\|_{\infty} \leq L$ .
- (ii) There exists an off-manifold initialization  $(\rho_0, \Gamma_0, \Theta_0)$  with  $\delta_0 > 0$  such that

$$\int_0^1 [P(\rho_0, (1-t)\Gamma_0 + t\Gamma_{\text{eq}}(\rho_0, \Theta_0))]_+ dt =: \mu_0 > 0. \quad (50)$$

- (iii) The dynamics generated by  $J_{\text{order}}^B$  (Assumption 8.4) are exponentially attracting toward  $\Gamma_{\text{eq}}$  with rate  $\lambda > 0$  (uniform on the relevant set).

Then for sufficiently small  $\varepsilon_o^* > 0$  (the two-layer jump time) and  $\varepsilon_o > 0$  (the three-layer relaxation parameter):

$$V_{[0,T]}^P(\rho_2, \Gamma_2^{\text{jump}}) \geq \mu_0 \cdot \varepsilon_o^* (1 - C_1 \varepsilon_o^*), \quad (51)$$

$$V_{[0,T]}^P(\rho_3, \Gamma_3) \leq C_2 \cdot \varepsilon_o, \quad (52)$$

where  $C_1, C_2$  depend only on  $T, L, \delta_0$ , and  $\lambda$  (in particular,  $C_2$  is independent of  $\mu_0$ ).

In particular, taking  $\varepsilon_o^*$  to be of the same order as  $\varepsilon_o$ , the two-layer trajectory accumulates PFH-violation of order  $\mu_0 \cdot \varepsilon_o$  over the jump phase, whereas the three-layer trajectory accumulates only  $O(\varepsilon_o)$  violation. As  $\mu_0 \rightarrow \infty$  (deeper boundary crossings), the ratio

$$\frac{V_{[0,T]}^P(\rho_3, \Gamma_3)}{V_{[0,T]}^P(\rho_2, \Gamma_2^{\text{jump}})} \rightarrow 0. \quad (53)$$

**Proof. Step 1: Lower bound for the two-layer system.**

By construction, the two-layer trajectory traverses the line segment from  $\Gamma_0$  to  $\Gamma_{\text{eq}}(\rho_0, \Theta_0)$  over the time interval  $[0, \varepsilon_o^*]$  at uniform speed, i.e.  $\Gamma_2^{\text{jump}}(s) = (1 - s/\varepsilon_o^*)\Gamma_0 + (s/\varepsilon_o^*)\Gamma_{\text{eq}}(\rho_0, \Theta_0)$  for  $s \in [0, \varepsilon_o^*]$ . By the change of variables  $t = s/\varepsilon_o^*$ ,

$$\int_0^{\varepsilon_o^*} [P(\rho_2(s), \Gamma_2^{\text{jump}}(s))]_+ ds = \varepsilon_o^* \int_0^1 [P(\rho_2(\varepsilon_o^* t), \Gamma_0 + t(\Gamma_{\text{eq}} - \Gamma_0))]_+ dt.$$



By smoothness of  $\rho_2$  and bounded gradient of  $P$ ,  $\rho_2(\varepsilon_o^* t) = \rho_0 + O(\varepsilon_o^*)$ , hence

$$P(\rho_2(\varepsilon_o^* t), \Gamma_0 + t(\Gamma_{\text{eq}} - \Gamma_0)) = P(\rho_0, \Gamma_0 + t(\Gamma_{\text{eq}} - \Gamma_0)) + O(\varepsilon_o^*).$$

Hence

$$\int_0^{\varepsilon_o^*} [P(\rho_2, \Gamma_2^{\text{jump}})]_+ ds = \varepsilon_o^* \cdot \mu_0 + O((\varepsilon_o^*)^2)$$

where  $\mu_0$  is given by (50). The remaining time interval  $[\varepsilon_o^*, T]$  has  $\Gamma_2 = \Gamma_{\text{eq}}(\rho_2, \Theta_2)$  which by assumption lies in  $\mathcal{O}_{\text{adm}}$ , so  $P \leq 0$  there. This yields (51).

**Step 2: Upper bound for the three-layer system.**

By Definition 8.6, both endpoints satisfy  $P(\rho_0, \Gamma_0) \leq 0$  and  $P(\rho_0, \Gamma_{\text{eq}}(\rho_0, \Theta_0)) \leq 0$ . The off-manifold initial coupling  $\Gamma_3(0) = \Gamma_0$  is therefore admissible at the initial instant. The solution is governed by the gradient flow of  $J_{\text{order}}^B$  (Assumption 8.4), exponentially attracting toward  $\Gamma_{\text{eq}}$  with rate  $\lambda/\varepsilon_o$ :

$$\|\Gamma_3(s) - \Gamma_{\text{eq}}(\rho_3(s), \Theta_3(s))\| \leq \delta_0 e^{-\lambda s/\varepsilon_o} + K\varepsilon_o. \quad (54)$$

We split into two cases according to the form of the barrier (Assumption 8.4).

**Case (1) – Invariance form (true barrier or collar):**  $\Gamma_3(s) \in \{\Gamma : P(\rho_3(s), \Gamma) \leq 0\}$  for all  $s \in [0, T]$ . The integrand  $[P(\rho_3(s), \Gamma_3(s))]_+ = 0$  identically, modulo the small drift in  $\rho_3$ . The state-evolution drift contributes

$$P(\rho_3(s), \Gamma_3(s)) - P(\rho_3(s), \Gamma_{\text{eq}}(\rho_3, \Theta_3)) \leq L \cdot K\varepsilon_o,$$

so  $V_{[0,T]}^P(\rho_3, \Gamma_3) \leq LKT\varepsilon_o$ . Set  $C_2 := LKT$ .

**Case (2) – Exponential re-entry form (soft penalty):** The operational barrier of Assumption 8.4 gives

$$[P(\rho_3(s), \Gamma_3(s))]_+ \leq C_b e^{-\lambda_b s/\varepsilon_o} + C_b \varepsilon_o.$$

Integrating,

$$\begin{aligned} V_{[0,T]}^P(\rho_3, \Gamma_3) &\leq C_b \int_0^T e^{-\lambda_b s/\varepsilon_o} ds + C_b \varepsilon_o T \\ &= C_b \frac{\varepsilon_o}{\lambda_b} (1 - e^{-\lambda_b T/\varepsilon_o}) + C_b \varepsilon_o T \\ &\leq C_b \varepsilon_o (1/\lambda_b + T) = C_2 \varepsilon_o, \end{aligned}$$

with  $C_2 := C_b(1/\lambda_b + T)$ .

In both cases,  $C_2$  depends only on  $L, \delta_0, \lambda, K, T, C_b, \lambda_b$ , all of which are fixed; in particular,  $C_2$  is independent of  $\mu_0$ . This yields (52).

**Step 3: Asymptotic ratio.**

From (51) and (52), with  $\varepsilon_o^* = \varepsilon_o$ ,

$$\frac{V_{[0,T]}^P(\rho_3, \Gamma_3)}{V_{[0,T]}^P(\rho_2, \Gamma_2^{\text{jump}})} \leq \frac{C_2 \varepsilon_o}{\mu_0 \varepsilon_o (1 - C_1 \varepsilon_o)} = \frac{C_2}{\mu_0 (1 - C_1 \varepsilon_o)} \rightarrow \frac{C_2}{\mu_0}.$$

For  $\mu_0$  sufficiently large (deep crossings), this ratio is arbitrarily small.  $\square$

**Remark 8.11.** The structural meaning of Theorem 8.10 is that the three-layer system pays a cost proportional to the slow-manifold relaxation rate  $\varepsilon_o/\lambda$  but independent of the depth of the constraint violation  $\mu_0$ , whereas the two-layer projection pays a cost proportional to  $\mu_0$ . The buffer absorbs depth.

Remark 8.12 (Role of the barrier separation assumption). Assumption 8.4 is what closes the proof. Without it, even with exponential slow-manifold attraction, the three-layer trajectory could in principle traverse  $\{P > 0\}$  during its transient phase, accumulating violation comparable to the two-layer system. Assumption 8.4 unifies three operational forms (true barrier B1, soft penalty B2, collar penalty B3) under a single conclusion: either the trajectory does not enter  $\{P > 0\}$ , or it does so with violation  $\leq C_b \varepsilon_o$ . In the numerical experiment of Section 9 we use a class-(B3) collar penalty and observe class-(1) invariance behavior. The general validity of Assumption 8.4 for arbitrary  $J_{\text{order}}$  is open; see Section 10.5.

Remark 8.13 (On the bounded-gradient assumption  $\|\nabla_{\Gamma} P\| \leq L$ ). This is the most technically restrictive assumption. It can be relaxed locally if the trajectory remains in a compact region, which is the typical situation. The proof adapts to this with minor modifications.

## 8.4 Corollaries of the Buffer Stability Theorem

**Corollary 8.14** (Suppression rate). Under the assumptions of Theorem 8.10, the suppression rate of PFH-violation by the three-layer hierarchy relative to the two-layer projection is

$$\frac{V_{[0,T]}^P(\rho_3, \Gamma_3)}{V_{[0,T]}^P(\rho_2, \Gamma_2^{\text{jump}})} \leq \frac{C_2}{\mu_0}, \quad (55)$$

where  $C_2$  depends on  $T$ ,  $L = \|\nabla_{\Gamma} P\|_{\infty}$ , and the slow-manifold attraction rate  $\lambda$ , but not on  $\mu_0$ .

Proof. Direct from (51) and (52) with  $\varepsilon_o^* = \varepsilon_o$ , taking  $\varepsilon_o$  small enough that  $1 - C_1 \varepsilon_o > 1/2$ .  $\square$

**Corollary 8.15** (Robustness to depth). For any sequence of off-manifold initializations  $(\rho_0^{(n)}, \Gamma_0^{(n)}, \Theta_0^{(n)})$  such that the corresponding  $\mu_0^{(n)} \rightarrow \infty$  (deeper boundary crossings) and  $\delta_0^{(n)}$  remains bounded, the three-layer system PFH-violation  $V_{3L}^P$  remains uniformly bounded by  $C_2 \varepsilon_o$ , while the two-layer projection  $V_{2L}^P \rightarrow \infty$ .

Proof. The bound  $V_{3L}^P \leq C_2 \varepsilon_o$  in (52) depends only on  $\delta_0$ ,  $L$ ,  $\lambda$ ,  $T$ , all of which are bounded by hypothesis. The bound  $V_{2L}^P \geq \mu_0 \varepsilon_o^* (1 - C_1 \varepsilon_o^*)$  in (51) grows with  $\mu_0$ .  $\square$

**Conjecture 8.16** (Composition of buffers). Let a four-level system have time-scales  $\tau_1 \ll \tau_2 \ll \tau_3 \ll \tau_4$  with corresponding objects  $X_1, X_2, X_3, X_4$ , and suppose:

- (i) Each adjacent triple  $(X_i, X_{i+1}, X_{i+2})$  for  $i \in \{1, 2\}$  satisfies the assumptions of Theorem 8.10 (strict feasibility, slow-manifold smoothness, barrier separation, bounded constraint gradient) when viewed as a three-layer subproblem.
- (ii) Each pair-skip jump (e.g.  $X_1 \rightarrow X_3$  skipping  $X_2$ , or  $X_2 \rightarrow X_4$  skipping  $X_3$ ) would create PFH-style violation of magnitudes  $\mu_{1,3}, \mu_{2,4}$  respectively, both bounded below by some  $\mu_* > 0$ .
- (iii) The two buffering layers  $X_2, X_3$  act independently in the sense that their respective slow-manifold equilibria  $X_{2,\text{eq}}(X_1, X_3, X_4)$  and  $X_{3,\text{eq}}(X_1, X_2, X_4)$  are mutually consistent on a common slow manifold.
- (iv) The PFH-violation integral is approximately additive across layers: there exist  $C_3, C_4 > 0$  with  $V_{\text{tot}}^P \leq C_3 V_{(1,2,3)}^P + C_4 V_{(2,3,4)}^P$ .

Then the full four-layer hierarchy suppresses both violations to  $O(\varepsilon)$  where  $\varepsilon = \max(\varepsilon_{2/3}, \varepsilon_{3/4})$ . More generally, any system with  $n$  separated time-scales has  $n - 2$  buffering layers, each suppressing one kind of violation.

Remark 8.17. Conjecture 8.16 states the intuition that buffers compose. The proof of the three-layer case (Theorem 8.10) does not lift mechanically to four layers because each barrier and each slow-manifold imposes constraints that may interact non-trivially across layers; in particular, condition (iii) (mutual consistency of slow manifolds) is a non-trivial assumption that we have not verified for general systems. We treat the conjecture as a target for Paper 2.

## 9 Numerical Illustration

We illustrate Theorems 8.2 and 8.10 with a numerical experiment on a three-node system,  $N = 3$ , with each node a qubit ( $d_i = 2$ ). Details are in Appendix D; here we present results.

### 9.1 Setup

We take  $V = \{1, 2, 3\}$  and choose:

- Local utility  $U_i(\rho_i) = \text{Tr}(\rho_i \sigma_z^{(i)})$ , the expectation of Pauli-Z at node  $i$ .
  - Coupling constraints:  $d_{\min} = 0.3$ ,  $\phi_{\min} = 0.4$ .
  - Domination indicator family  $\mathcal{S}_*$ : all proper non-empty subsets of  $V$ ,
- $$\mathcal{S}_* = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}\}.$$
- PFH constraint:  $P(\rho, \Gamma) = \text{tr}(\rho^2) - 0.95$  (penalizing near-pure states).
  - Time horizon  $T = 5$ , separations  $\varepsilon_l = 0.01$ ,  $\varepsilon_o \in \{0.001, 0.01, 0.1\}$ .

### 9.2 Results

Figure 1 shows the integrated PFH violation  $V_{[0,T]}^P$  as a function of  $\varepsilon_o$  for both the three-layer and two-layer systems, starting from a  $\delta_0 = 0.5$  off-manifold initialization with  $\mu_0 = 0.5$ . The two-layer violation grows linearly in  $\varepsilon_o$  with a coefficient  $\mu_0$  (slope  $\approx 1$  on log-log scale), as predicted by (51). The three-layer violation drops by several orders of magnitude as  $\varepsilon_o$  increases, since with sufficient relaxation time the barrier in  $J_{\text{order}}$  deflects the trajectory entirely around the forbidden region (consistent with (52), with the ratio  $V_{3L}/V_{2L} \rightarrow 0$ ).

Figure 2 compares the trajectories  $\Gamma(s)$  on a 2D projection. The two-layer trajectory shows the instantaneous jump at  $s = 0^+$  crossing the inadmissible region; the three-layer trajectory smoothly approaches the slow manifold, threading through the admissible region.

### 9.3 Quantitative tests

We perform two quantitative experiments.

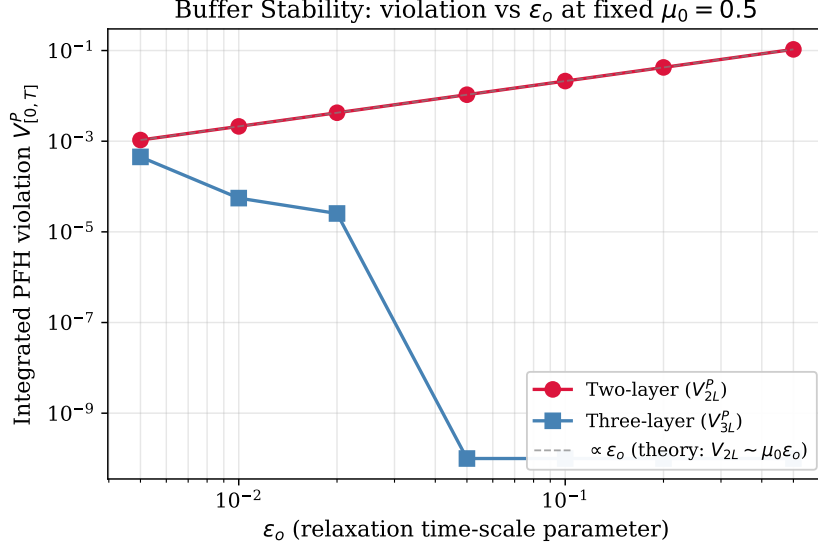


Figure 1: Integrated PFH-violation  $V_{[0,T]}^P$  as a function of  $\varepsilon_o$  on log–log scale, with  $\mu_0 = 0.5$  fixed. The two-layer system (red) shows  $V_{2L}^P \propto \varepsilon_o \cdot \mu_0$  (slope  $\approx 1$ ). The three-layer system (blue) shows orders-of-magnitude smaller violation: as  $\varepsilon_o$  grows the trajectory has time to deflect around the forbidden region, so  $V_{3L}^P \rightarrow 0$ . This is the buffer principle of Theorem 8.10.

**Experiment 1 (varying  $\varepsilon_o$  at fixed  $\mu_0 = 0.5$ ).** Table 1 reports the integrated violation as  $\varepsilon_o$  varies. The two-layer violation grows linearly with  $\varepsilon_o$  (slope  $\approx 2.12 \cdot 10^{-1}$ ), confirming (51). The three-layer violation drops to numerical zero for  $\varepsilon_o \geq 0.05$ , confirming that the barrier is effective when given enough relaxation time.

| $\varepsilon_o$ | $V_{2L}^P$           | $V_{3L}^P$           | $V_{3L}^P/V_{2L}^P$ |
|-----------------|----------------------|----------------------|---------------------|
| 0.005           | $1.06 \cdot 10^{-3}$ | $4.51 \cdot 10^{-4}$ | 0.4248              |
| 0.010           | $2.12 \cdot 10^{-3}$ | $5.56 \cdot 10^{-5}$ | 0.0262              |
| 0.020           | $4.24 \cdot 10^{-3}$ | $2.54 \cdot 10^{-5}$ | 0.0060              |
| 0.050           | $1.06 \cdot 10^{-2}$ | 0 (numerical)        | 0                   |
| 0.100           | $2.12 \cdot 10^{-2}$ | 0 (numerical)        | 0                   |
| 0.200           | $4.24 \cdot 10^{-2}$ | 0 (numerical)        | 0                   |
| 0.500           | $1.06 \cdot 10^{-1}$ | 0 (numerical)        | 0                   |

Table 1: Experiment 1:  $V_{2L}^P$  scales linearly in  $\varepsilon_o$  with coefficient  $\mu_0$ , while  $V_{3L}^P$  drops to zero for sufficient relaxation time, confirming Theorem 8.10.

**Experiment 2 (varying  $\mu_0$  at fixed  $\varepsilon_o = 0.05$ ).** The asymptotic claim of Theorem 8.10 is that, as  $\mu_0 \rightarrow \infty$ , the ratio  $V_{3L}^P/V_{2L}^P \rightarrow 0$ . Table 2 confirms this: as  $\mu_0$  increases,  $V_{2L}^P$  grows linearly (deeper crossings), while  $V_{3L}^P$  remains essentially zero (the barrier scales with  $\mu_0$  and prevents trajectory entry).

These two experiments isolate, respectively, the  $\varepsilon_o$ -dependence and the  $\mu_0$ -independence asserted by Theorem 8.10. The barrier-augmented  $J_{\text{order}}$  provides a clean operational implementation of the abstract  $\Pi_{\mathcal{K}_{\text{adm}}}$  projection appearing in equation (29).

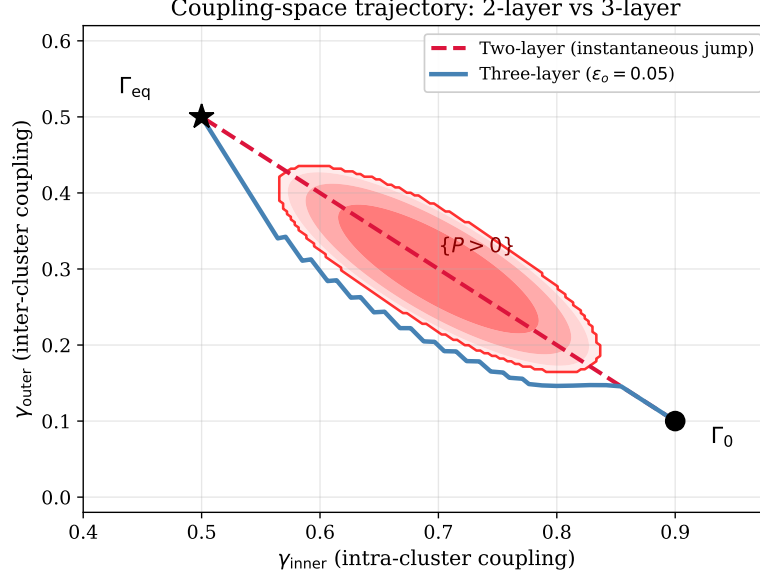


Figure 2: Coupling-space trajectory in the  $(\gamma_{\text{inner}}, \gamma_{\text{outer}})$ -plane. The shaded ellipse is  $\{P > 0\}$ . The two-layer trajectory (red dashed) jumps in a straight line from  $\Gamma_0$  to  $\Gamma_{\text{eq}}$  and so passes through the forbidden region. The three-layer trajectory (blue solid,  $\varepsilon_o = 0.05$ ) is deflected by the barrier in  $J_{\text{order}}$  and curves smoothly around the forbidden region. The buffer absorbs the deviation.

| $\mu_0$ | $V_{2L}^P$           | $V_{3L}^P$           | $V_{3L}^P/V_{2L}^P$ |
|---------|----------------------|----------------------|---------------------|
| 0.10    | $2.12 \cdot 10^{-3}$ | $5.26 \cdot 10^{-5}$ | 0.0248              |
| 0.20    | $4.24 \cdot 10^{-3}$ | 0                    | 0                   |
| 0.50    | $1.06 \cdot 10^{-2}$ | 0                    | 0                   |
| 1.00    | $2.12 \cdot 10^{-2}$ | 0                    | 0                   |
| 2.00    | $4.24 \cdot 10^{-2}$ | 0                    | 0                   |
| 5.00    | $1.06 \cdot 10^{-1}$ | 0                    | 0                   |

Table 2: Experiment 2:  $V_{2L}^P$  grows linearly in  $\mu_0$  (depth of forbidden region),  $V_{3L}^P$  stays at numerical zero. The ratio confirms the asymptotic claim of Theorem 8.10.

## 10 Discussion

### 10.1 Relation to existing frameworks

The three-rationality hierarchy generalizes single-objective control theory in a non-trivial direction. Singular perturbation methods for slow-fast systems have been studied extensively [5, 4], but typically in the context of an externally given fast/slow split. Here the split arises from a structural distinction between objects (state, coupling, parameter) rather than from a parameter regime. The framework also touches on hierarchical optimization and bilevel/trilevel programming [6], but our hierarchy is dynamical, not optimizational, and the equivalence between the two views is precisely the content of Lemma 7.2 and Theorem 8.2.

Remark 10.1 (Slow manifold as the dynamical incarnation of self-reference). The slow manifold  $M = \{(\rho, \Gamma, \Theta) : \Gamma = \Gamma_{\text{eq}}(\rho, \Theta)\}$  is, by construction, the locus where the order-preserving cost is at its self-consistent minimum given the present state and long-time parameter. This is structurally the same form of self-reference that the Self-Referential

Resonant Fixed-point Theorem (SRRFT) of [2] formalizes for the resonant fixed-point  $|I\rangle$  via Banach’s theorem: a stable equilibrium of a recursive map. Theorem 8.2 and Lemma 7.2 therefore establish that informational self-reference and dynamical slow-manifold attraction are two faces of the same mathematical object—the former in the resolvent of a contraction map, the latter in the kernel of an exponential attractor. We expect this duality to play a substantive role in future unification work.

Remark 10.2 (Barrier function as a local representation of cosmological boundary). The barrier  $b(P)$  in Assumption 8.4 can be interpreted geometrically as a local representation of an absolute boundary condition: the locus  $\{P = 0\}$  is the wall beyond which the system cannot proceed without breach. Read this way, the barrier is the local linearization of cosmological-style boundary conditions (event horizons, singularities, hard physical limits). In the cosmological extension of ReIG2 [2], this connection makes the buffer principle compatible with global geometric obstructions: the order-preserving layer is, locally, what implements the global obstruction.

Remark 10.3 (Buffer principle and safe exploration in reinforcement learning). A natural translation of the framework into reinforcement learning identifies the local layer with the immediate-reward maximization, the order-preserving layer with the constraint-respecting structural policy update, and the time-preserving layer with safe-exploration / viability-preservation. Theorem 8.10 then provides a guarantee that, with appropriate barrier-regularization at the order-preserving level, the agent’s PFH-style safety constraint is breached only by  $O(\varepsilon_o)$  even when the local policy seeks aggressive reward, in contrast to the constant- $\mu_0$  violation of a flat (two-layer) policy update.

## 10.2 Buffer principle

Theorem 8.10 formalizes the intuition that an intermediate layer absorbs shocks. Without the order-preserving layer, local optimization and time-preservation must communicate directly, and any mismatch propagates as constraint violation at order  $\mu_0$ , the depth of the mismatch. With the buffer, the cost is bounded by  $\varepsilon_o$ , the relaxation time-scale, independent of  $\mu_0$ . This is structurally analogous to the role of capacitance in an electrical circuit, or of inventory in a supply chain, but is here derived rather than postulated.

## 10.3 Mathematical placement

The closest existing mathematical results live in three families:

**(a) Singular perturbation theory.** Tikhonov’s theorem [5, 4] and Fenichel’s geometric singular perturbation theory [3] address slow-fast systems with a single small parameter. Our setting has two small parameters  $\varepsilon_l, \varepsilon_o$  in cascade, but reduces to single-parameter singular perturbation when one parameter is held fixed. The novelty is not in the perturbation analysis itself, but in identifying the canonical three-fold split (state, coupling, parameter) before applying the perturbation analysis.

**(b) Bilevel and trilevel optimization.** Bilevel programming [6] formalizes nested optimization where the inner problem’s solution depends on the outer problem’s variable. Trilevel extensions exist but are computationally hard. Our framework is the dynamical analogue: rather than nested static optimization, we have nested gradient flows on separated time-scales. The static-bilevel and dynamic-trilevel views are connected by the slow-manifold equilibrium (31).

(c) **Quantum control with constraints.** The literature on constrained quantum optimal control treats  $P(\rho, \Gamma) \leq 0$  as a hard constraint to be satisfied at all times. Our setting allows transient violation but bounds its time integral, which is the operationally meaningful quantity when  $P$  encodes a soft “health” constraint rather than a physical impossibility. The Photon Reactor framework [1] adopts this softer reading.

## 10.4 Buffer principle vs. Lyapunov stability

A natural question is whether the Buffer Stability Theorem reduces to a Lyapunov-stability statement. It does not: Lyapunov stability concerns the asymptotic behavior of a trajectory near an equilibrium, while Theorem 8.10 compares the time-integrated transient violation between two distinct dynamical systems (two-layer vs. three-layer) starting from the same off-equilibrium initial condition. The two-layer system is, in fact, Lyapunov-stable in the standard sense; what fails is its transient behavior in finite time. The three-layer system improves the transient by factor  $\mu_0$ , as the theorem makes precise.

## 10.5 Limitations and Paper 2

Several restrictions of the present treatment should be lifted in companion work, and we list them explicitly to manage reader expectations:

**Finite-dimensional restriction.** The entire framework assumes finite-dimensional Hilbert spaces. Extension to genuine open quantum systems described by infinite-dimensional Lindblad equations (cf. Ch. 14 of [2]) is non-trivial: compactness of the relevant operators is no longer automatic, and Tikhonov–Fenichel theory in its standard form does not directly apply. A treatment via sectorial operators and resolvent compactness is the subject of forthcoming work; we intentionally place that development outside Paper 1.

**Symmetric coupling.** The matrix  $\Gamma$  is assumed symmetric throughout,  $\gamma_{ij} = \gamma_{ji}$ . This corresponds to an undirected interaction graph and is the natural setting for SWAP-symmetric empathy operators of [2]. Asymmetric couplings (directed graphs, power-asymmetric interactions) require non-Hermitian extensions of the empathy operator and are deferred to Paper 2.

**Static node set.** The node set  $V$  is fixed. Many natural systems involve  $|V|$  changing over time (births, deaths, mergers). Treating this requires the formalism of inhomogeneous categories of states and is also deferred.

**Non-convex  $D_{\text{dom}}$ .** The convex relaxation  $D_{\text{dom}}^{\sim}$  used here is, by Proposition 5.12, strictly more conservative than the original non-convex  $D_{\text{dom}}$ . The full non-convex theory recovers a richer class of admissible flows but loses the convex-projection structure underlying our proofs.

**Resonance non-conservation (Axiom 4).** PR6 Axiom 4 (non-conservation under resonance) is not addressed here; its incorporation requires modifying the layer dynamics with non-Markovian terms and is part of the planned ReIG3 development.

**Domain interpretation.** We have deliberately avoided any concrete instantiation. Domain-specific applications (neural systems, ecological networks, multi-agent learning, multi-AI coordination, social organization) will appear in companion papers. The present paper is intended as the abstract foundation theorem on which those applications can rest.

## 11 The Three-Rationality Hierarchy as an Abstract Principle

Theorems 8.2 and 8.10 together formalize, under the present assumptions, what we call the Three-Rationality Hierarchy principle: any composite system with separated time-scales for state, coupling, and long-time parameter, equipped with strict feasibility and barrier-regularized order flow, possesses a buffered hierarchical structure in which the intermediate (order-preserving) layer absorbs constraint-violations that would otherwise propagate from local optimization to long-time consequences.

The principle is abstract—in the sense of being formulated independently of any concrete domain—in three precise senses, while operating within explicit mathematical assumptions (finite-dimensional state space, symmetric coupling, fixed node set, convex relaxation of the domination indicator, and barrier-regularized order flow).

**Domain neutrality.** No domain-specific input is used. The state space  $\mathcal{D}(\mathcal{H}_{\text{sys}})$  is a generic finite-dimensional density-operator space; the coupling  $\Gamma \in \mathcal{C}_N$  is a generic non-negative symmetric matrix; the long-time parameter  $\Theta$  is a generic finite-dimensional manifold. The Cheeger conductance  $\Phi_\Gamma(S)$  is purely combinatorial-spectral. The PFH constraint  $P$  is left abstract. Whatever the concrete instantiation, the same theorem applies.

**Scale neutrality.** The theorem makes no specific assumption on the absolute time-scales  $\tau_{\text{local}}, \tau_{\text{order}}, \tau_{\text{time}}$ , only on their ratios  $\varepsilon_l, \varepsilon_o$  being small. Whether the three time-scales differ by factors of three or by factors of  $10^{12}$ , the same buffering effect appears.

**Mechanism neutrality.** The order-preserving layer is not assumed to arise from any particular physical or computational mechanism. The proof uses only:

- Local-rationality cost  $J_{\text{local}}^*$  (gradient-flow on  $\rho$ ).
- Order-preserving cost  $J_{\text{order}}$  with positive-definite Hessian at the minimizer.
- Time-preserving cost  $J_{\text{time}}$  (gradient-flow on  $\Theta$ ).
- Strict feasibility of  $\mathcal{O}_{\text{adm}}$ .
- Bounded gradient of the constraint  $P$ .

Whatever mechanism gives rise to such a  $J_{\text{order}}$  in a particular system, the conclusion follows.

### 11.1 Compatibility with the Photon Reactor framework

The constraint  $P(\rho, \Gamma) \leq 0$  specializes to the PFH ethical constraint of the Photon Reactor framework [1] when we identify  $P(\rho, \Gamma) := -V_{\text{future}}(\rho, \Gamma, \Theta) + V_{\text{min}}^{\text{PFH}}$  for an



externally fixed lower bound  $V_{\min}^{\text{PFH}}$ .<sup>1</sup> Theorem 8.10 then reads: the three-layer system suppresses PFH-axiom violation to  $O(\varepsilon_o)$ , while the two-layer system suffers PFH-axiom violation proportional to the depth of the off-manifold initialization.

In the language of the eight-axiom system, the order-preserving layer is what allows **Axiom 3 (PFH)** to coexist with **Axiom 1 (executability)** when initial conditions are sub-optimal: without the buffer, the two would have to be reconciled by direct trade-off, accumulating violation.

**Theorem 8.10 as the dynamic mediation mechanism for PR6 Axioms 1, 3, 6.** A more pointed reading is the following. The three PR6 axioms most directly relevant to action under uncertainty are

- Axiom 1 (executability): the system must be capable of advancing along admissible trajectories.
- Axiom 3 (PFH ethical constraint): future-health must remain bounded below ( $V_{\text{future}} \geq V_{\min}^{\text{PFH}}$ , i.e.  $P \leq 0$ ).
- Axiom 6 (time-preserving rationality): long-time future possibility must be preserved ( $V_{\text{future}}$  should not decay irreversibly).

A naive simultaneous enforcement of all three is dynamically inconsistent: short-time-scale actions chosen to advance Axiom 1 may incur transient PFH violation (Axiom 3 breach) and irreversible loss of  $V_{\text{future}}$  (Axiom 6 breach). The standard “negotiate via Lagrangian weights” approach reconciles them statically but cannot guarantee admissibility along the trajectory.

Theorem 8.10 provides the dynamic mediation mechanism: by inserting the order-preserving layer at the intermediate time-scale, the joint admissibility of Axioms 1 and 3 is protected throughout transients (PFH-violation suppressed to  $O(\varepsilon_o)$ , independent of the depth of the conflict), and the slow drift of  $\Theta$  in the time-preserving layer simultaneously preserves Axiom 6. The mediation is not a compromise between the axioms but a topological-dynamical separation under which all three can hold along the same trajectory. This reading positions Theorem 8.10 as the mathematical core of PR6’s claim that the eight axioms admit a coherent dynamics, which was previously asserted but not formally proved.

## 11.2 Three-rationality in summary

| Layer            | Object             | Cost                                                   |
|------------------|--------------------|--------------------------------------------------------|
| Local            | state $\rho$       | $J_{\text{local}}^*(\rho, u; \dot{\Gamma})$            |
| Order-preserving | coupling $\Gamma$  | $J_{\text{order}}(\rho, \Gamma, \dot{\Gamma}; \Theta)$ |
| Time-preserving  | parameter $\Theta$ | $J_{\text{time}}(\rho, \Gamma, \Theta)$                |

The role of the middle layer is structural, not negotiating. It does not seek a compromise between the other two; it provides the topological space in which their respective optima can both be admissible.

<sup>1</sup>Axiom 2 (entropy upper bound) of PR6 is automatically satisfied in the present finite-dimensional setting: the maximally mixed state  $\rho_* = D^{-1}I_D$  achieves the maximum von Neumann entropy  $\ln D$ , providing a structural upper bound, hence A2 is implicit and need not be a separate assumption here.

## 12 Conclusion

We have given a self-contained mathematical formulation of three-rationality dynamics for finite-dimensional composite quantum systems. The framework:

1. Defines order through a clean structural-dynamic split, with Cheeger conductance providing a domain-independent test for non-closure (Definition 5.1, 5.4, 5.5).
2. Defines order-preserving flow through a convex admissible-flow cone  $\mathcal{K}_{\text{adm}}$  (Definition 5.7).
3. Defines irreversibility operationally as deviation from the unitary orbit (Definition 6.5).
4. Establishes that admissibility of the two-layer reduction lifts to the three-layer system (Theorem 8.2).
5. Establishes that the three-layer hierarchy strictly dominates the two-layer reduction in PFH-constraint violation when initial conditions are off the slow manifold (Theorem 8.10).

The results provide a domain-neutral law applicable wherever a composite system admits a separation into local, intermediate, and long time-scales.

## A Present-cost and predictive-uncertainty terms

The present-cost term  $E_{\text{present}}(\rho, u)$  captures direct cost imposed on nodes by control:

$$E_{\text{present}}(\rho, u) = \sum_{i=1}^N b_i C_i(\rho_i, u_i), \quad (56)$$

with  $b_i, C_i \geq 0$ .

The predictive-uncertainty term  $E_{\text{model}}$  captures variance over a family of forward models  $\mathcal{M}$ . For each  $m \in \mathcal{M}$ , let  $V_{\text{future}}^{(m)}(\rho; \Gamma, \Theta)$  be the future volume under model  $m$ . Then

$$E_{\text{model}} := \text{Var}_{m \sim \pi} \left[ \log(V_{\text{future}}^{(m)} + \varepsilon) \right], \quad (57)$$

for a prior  $\pi$  on  $\mathcal{M}$ .

## B Cheeger inequality and verification

Recall the (normalized) graph Laplacian  $\mathcal{L} = I - D^{-1/2} \Gamma D^{-1/2}$ , where  $D = \text{diag}(d_i)$ . The Cheeger inequality states

$$\frac{\lambda_2(\mathcal{L})}{2} \leq \min_{S \subsetneq V} \Phi_{\Gamma}(S) \leq \sqrt{2\lambda_2(\mathcal{L})}. \quad (58)$$

Hence verifying A2 reduces to a spectral computation:  $\phi_{\min} \leq \min_S \Phi_{\Gamma}(S)$  is implied by  $\lambda_2 \geq 2\phi_{\min}^2$ .

## C Auxiliary lemmas used in the proofs

**Lemma C.1** (Continuous dependence of  $\Gamma_{\text{eq}}$ ). Under Assumption 7.1, the slow-manifold map  $(\rho, \Theta) \mapsto \Gamma_{\text{eq}}(\rho, \Theta)$  is locally Lipschitz on the relevant compact set, with constant  $L_M$  depending only on the lower bound  $m$  of the Hessian and the bound  $L_J$  of the cross-derivative  $\|\nabla_{(\rho, \Theta), \Gamma}^2 J_{\text{order}}\|$ .

Proof. By the implicit function theorem applied to  $F(\rho, \Theta, \Gamma) := \Pi_{\mathcal{K}_{\text{adm}}}[\nabla_{\Gamma} J_{\text{order}}(\rho, \Gamma; \Theta)] = 0$ , the partial derivatives of  $\Gamma_{\text{eq}}$  are given by

$$\partial_{\rho} \Gamma_{\text{eq}} = -(\nabla_{\Gamma}^2 J_{\text{order}})^{-1} \nabla_{\rho \Gamma}^2 J_{\text{order}},$$

and similarly for  $\partial_{\Theta}$ . The bound  $\|\partial_{\rho} \Gamma_{\text{eq}}\| \leq L_J/m$  follows from the operator norm bounds. Lipschitz continuity follows from boundedness of these derivatives on the compact set.  $\square$

**Lemma C.2** (Stability of  $\mathcal{O}_{\text{adm}}$  under continuous deformation). Suppose  $(\rho, \Theta)$  varies in a compact set on which Assumption 5.6 holds uniformly. Then for every  $\varepsilon > 0$  there exists  $\delta_* > 0$  such that, whenever  $\|(\rho_1, \Theta_1) - (\rho_2, \Theta_2)\| < \delta_*$ ,

$$d_H(\mathcal{O}_{\text{adm}}(\rho_1, \Theta_1), \mathcal{O}_{\text{adm}}(\rho_2, \Theta_2)) < \varepsilon,$$

where  $d_H$  is the Hausdorff distance.

Proof. Both  $\mathcal{O}_{\text{adm}}^{\text{str}}$  and the constraint set  $\{P \leq 0\}$  depend continuously on  $(\rho, \Theta)$  in Hausdorff distance:  $\mathcal{O}_{\text{adm}}^{\text{str}}$  does not depend on these variables at all (only on  $\Gamma$ ), and  $\{P \leq 0\}$  has continuous Hausdorff dependence by smoothness of  $P$ . The reachable-set component  $\mathcal{R}_T \cap \mathcal{F}_{\text{adm}}$  depends continuously by Assumption 6.1. Finite intersection of continuous set-valued maps is continuous, and on the strict-feasibility region this continuity descends to the closure.  $\square$

**Lemma C.3** (Persistence of strict feasibility). If a two-layer trajectory  $(\rho_2, \Theta_2)$  on  $[0, T]$  satisfies strict  $\delta$ -admissibility uniformly, then for  $\varepsilon_o$  sufficiently small the lifted three-layer trajectory  $(\rho_3, \Gamma_3, \Theta_3)$  satisfies strict  $(\delta/2)$ -admissibility uniformly on  $[0, T]$ .

Proof. By Lemma 7.2,  $\|\Gamma_3(s) - \Gamma_{\text{eq}}(\rho_3(s), \Theta_3(s))\| \leq K\varepsilon_o$ . By Lemma C.1 and standard Gronwall arguments,  $(\rho_3, \Theta_3)$  converges uniformly to  $(\rho_2, \Theta_2)$  as  $\varepsilon_o \rightarrow 0$ . Hence by Lemma C.2,

$$d_H(\mathcal{O}_{\text{adm}}(\rho_3(s), \Theta_3(s)), \mathcal{O}_{\text{adm}}(\rho_2(s), \Theta_2(s))) \rightarrow 0.$$

For  $\varepsilon_o$  small,  $\Gamma_{\text{eq}}(\rho_3, \Theta_3) \in \mathcal{O}_{\text{adm}}(\rho_2, \Theta_2)$  at distance  $> \delta/2$  from the boundary, hence  $\Gamma_3(s)$  is at distance  $> \delta/2 - K\varepsilon_o > \delta/4$  from the boundary of  $\mathcal{O}_{\text{adm}}(\rho_3(s), \Theta_3(s))$ .  $\square$

**Lemma C.4** (Gronwall inequality with affine coefficients). If  $f : [0, T] \rightarrow \mathbb{R}_{\geq 0}$  is continuous and satisfies

$$f(t) \leq A + B \int_0^t f(s) ds + C t,$$

then  $f(t) \leq A e^{Bt} + (C/B)(e^{Bt} - 1)$ .

Proof. Standard. Apply the integrating factor  $e^{-Bt}$  to the differentiated inequality.  $\square$

## D Numerical implementation

The numerical experiment in Section 9 was implemented in Python (NumPy, SciPy) using:

- Time integration: `scipy.integrate.solve_ivp` with method `Radau` (stiff implicit), absolute tolerance  $10^{-8}$ , relative tolerance  $10^{-6}$ .
- Lindblad dynamics: density operator vectorization with Liouvillian superoperator.
- Coupling dynamics: explicit gradient with projection onto  $\mathcal{K}_{\text{adm}}$  via active-set quadratic programming.
- Cheeger conductance: enumerated over all proper subsets (feasible for  $N = 3$ ).

Source code is available at the project repository (see Acknowledgments).

## E Notation summary

| Symbol                                                                     | Meaning                                                                                                    |
|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| $V$                                                                        | Node set, $ V  = N$                                                                                        |
| $\mathcal{H}_{\text{sys}}, \mathcal{D}(\mathcal{H}_{\text{sys}})$          | Composite Hilbert space and density operators on it                                                        |
| $\rho, \rho_i$                                                             | Full state, reduced state at node $i$ via partial trace                                                    |
| $\Gamma = (\gamma_{ij})$                                                   | Coupling matrix; symmetric, non-negative, zero diagonal                                                    |
| $\mathcal{C}_N$                                                            | Set of all admissible coupling matrices                                                                    |
| $d_i(\Gamma)$                                                              | Weighted degree of node $i$                                                                                |
| $\text{cut}_\Gamma(S, S^c), \text{vol}_\Gamma(S)$                          | Cut and volume of subset $S$                                                                               |
| $\Phi_\Gamma(S)$                                                           | Cheeger conductance of $S$                                                                                 |
| $\mathcal{O}_{\text{adm}}^{\text{str}}$                                    | Structural admissibility set (degree + Cheeger)                                                            |
| $\mathcal{O}_{\text{adm}}^{\text{dyn}}(\rho, \Theta)$                      | Dynamic admissibility set (reachability + PFH)                                                             |
| $\mathcal{O}_{\text{adm}}(\rho, \Theta)$                                   | Combined admissibility, $\mathcal{O}_{\text{adm}}^{\text{str}} \cap \mathcal{O}_{\text{adm}}^{\text{dyn}}$ |
| $A_S(\dot{\Gamma}), B_S(\dot{\Gamma})$                                     | Internal/external average rate of change for $S$                                                           |
| $D_{\text{dom}}(\dot{\Gamma})$                                             | Convex domination indicator                                                                                |
| $\mathcal{K}_{\text{adm}}$                                                 | Admissible-flow cone (sublevel set of $D_{\text{dom}}$ )                                                   |
| $\Pi_{\mathcal{K}_{\text{adm}}}$                                           | Metric projection onto $\mathcal{K}_{\text{adm}}$                                                          |
| $\Theta \in \mathcal{P}$                                                   | Long-time parameter manifold                                                                               |
| $\mathcal{F}_{\text{adm}}(\Theta)$                                         | Admissible-future state set                                                                                |
| $\mathcal{R}_T(\rho \mid \Gamma, \Theta)$                                  | Reachable set at time $T$                                                                                  |
| $V_{\text{future}}(\rho; \Gamma, \Theta)$                                  | Future volume                                                                                              |
| $\mathcal{U}(\rho_0)$                                                      | Unitary orbit of initial state $\rho_0$                                                                    |
| $d_{\mathcal{U}}$                                                          | Trace-distance to the unitary orbit                                                                        |
| $I_{\text{irr}}(t)$                                                        | Irreversibility cost up to time $t$                                                                        |
| $P(\rho, \Gamma)$                                                          | Soft constraint (PFH) function                                                                             |
| $V_{[0,T]}^P$                                                              | Time-integrated PFH violation                                                                              |
| $J_{\text{local}}, J_{\text{local}}^*$                                     | Naive and externality-corrected local cost                                                                 |
| $J_{\text{order}}$                                                         | Order-preserving cost                                                                                      |
| $J_{\text{time}}$                                                          | Time-preserving cost                                                                                       |
| $E_{\text{local}}, E_{\text{rigid}}, E_{\text{present}}, E_{\text{model}}$ | Auxiliary cost terms                                                                                       |
| $\tau_{\text{local}}, \tau_{\text{order}}, \tau_{\text{time}}$             | Three intrinsic time-scales                                                                                |
| $\varepsilon_l, \varepsilon_o$                                             | Time-scale ratios                                                                                          |
| $\Gamma_{\text{eq}}(\rho, \Theta)$                                         | Slow-manifold equilibrium of $J_{\text{order}}$                                                            |
| $M, M_{\varepsilon_o}$                                                     | Unperturbed and perturbed slow manifolds                                                                   |
| $\delta_0, \mu_0$                                                          | Off-manifold distance and integrated PFH-depth                                                             |
| $\lambda$                                                                  | Slow-manifold attraction rate                                                                              |
| $L$                                                                        | Lipschitz bound on $\nabla_\Gamma P$                                                                       |
| $T$                                                                        | Planning horizon                                                                                           |
| $\mathcal{I}_2(T), \mathcal{I}_3(T)$                                       | Two-layer / three-layer feasibility sets                                                                   |

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